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Locally Convex Topological Vector Spaces

Definitions and Examples

Definition: A topological vector space (TVS) is a vector space X with a topology $\mathcal{T}$ such that, with respect to this topology, the maps

(i) $X \times X \to X$, $(x, y) \mapsto x + y$

(ii) $\mathbb{F} \times X \to X$, $(\alpha, x) \mapsto \alpha x$

are continuous with respect to $\mathcal{T}$.

Example: If $\mathcal{P}$ is a family of seminorms on X, and $\mathcal{T}$ is the topology on X with subbasic sets $U(x_0, p, \varepsilon) = \{x \in X : p(x - x_0) < \varepsilon\}$, then $(X, \mathcal{T})$ is a topological vector space.

Definition: A locally convex topological vector space (LCS) is a TVS equipped with a family of seminorms $\mathcal{P}$ that satisfy

$$\bigcap_{p \in \mathcal{P}} \{x : p(x) = 0\} = \{0\}.$$

That is, the family $\mathcal{P}$ separates the points of X.

Proposition: Let X be a TVS, and let p be a seminorm on X. The following are equivalent:

(a) p is continuous;

(b) $\{x \in X : p(x) < 1\}$ is open;

(c) $0 \in \{x \in X : p(x) < 1\}^\circ$;

(d) $0 \in \{x \in X : p(x) \leq 1\}^\circ$;

(e) p is continuous at 0;

(f) there is a continuous seminorm q on X such that $p \leq q$.

Proof. We see immediately that (a) implies (b) since the set is equal to the inverse image of the open subset $[0,1) \subseteq [0,\infty)$. Similarly, if $p^{-1}([0,1))$ is open, then it is equal to its own interior, and since $0 \in p^{-1}([0,1))$, it follows that there is a subbasic open neighborhood of 0 contained in $p^{-1}([0,1))$, giving (c). Similarly, we obtain (d) by the fact that $0 \in p^{-1}([0,1))^\circ \subseteq p^{-1}([0,1])^\circ$.

Next, we see that (d) implies that for every $\varepsilon > 0$, $0 \in p^{-1}([0,\varepsilon])^\circ$, so if $(x_i)_i$ is a net in X converging to 0 and $\varepsilon > 0$, there is some i_0 such that for all $i \geq i_0$, $x_i \in \{x : p(x) \leq \varepsilon\}$. That is, $p(x_i) \leq \varepsilon$ for all $i \geq i_0$, whence p is continuous at 0. This gives (e).

If p is continuous at 0, then if $x_i \to x$ is a convergent net in X, the reverse triangle inequality gives

$$|p(x) - p(x_i)| \leq p(x_i - x),$$

whence $p(x_i) \to p(x)$, so p is continuous. This gives that (e) implies (a).

Taking $q = p$ yields (a) implies (f). We will show that (f) implies (e).If $x_i \to 0$ in X, then $q(x_i) \to 0$, but since $0 \leq p(x_i) \leq q(x_i)$, it follows that $p(x_i) \to 0$. □

Example:

(a) Let X be a completely regular topological space, and let $C(X)$ be the set of all continuous functions from X to $\mathbb{F}$. If K is compact in X, we may define a seminorm $p_K(f) = \sup_{x \in K} |f(x)|$. Then,

$$\{p_K : K \subseteq X \text{ compact}\}$$

forms a separating family of seminorms for $C(X)$.

(b) Let G be an open subset of $\mathbb{C}$. If $H(G)$ is the set of holomorphic functions on G, then the same seminorms $\{p_K : K \subseteq G \text{ compact}\}$ forms a locally compact space whose topology is uniform convergence on compact subsets.

(c) Let X be a normed vector space. For each $\varphi \in X^*$, let $p_\varphi(x) = \left|\varphi(x)\right|$. Then, $\mathcal{P} = \left\{p_\varphi : \varphi \in X^*\right\}$ forms a separating family of seminorms for X. The topology these seminorms induce is called the *weak topology* on X. It is often denoted $\sigma(X, X^*)$.

(d) If X is a normed space, then for each $x \in X$, we may define a seminorm $p_x \colon X^* \to [0,\infty)$ by taking $p_x(\varphi) = \left|\varphi(x)\right|$. This defines a separating family of seminorms $\mathcal{P} = \{p_x : x \in X\}$, and the topology on X^* induced by these seminorms is called the weak*, or w^*, topology. It is often denoted $\sigma(X^*, X)$.

Next, we focus on an examination of the "convexity" part of the notion of locally convex topological vector spaces.

Definition: A subset $A \subseteq X$ is said to be convex if for all $v, w \in A$, the segment

$$[v,w] := \{tv + (1-t)q : 0 \leq t \leq 1\}$$

is contained in A.

Note that the intersection of any family of convex subsets is convex.

Definition: If $A \subseteq X$ is any subset, then the *convex hull* of A, denoted $\text{conv}(A)$, is the intersection of all the convex subsets that contain A. If X is a TVS, the *closed convex hull* of A, denoted $\overline{\text{conv}}(A)$, is the intersection of all the closed convex sets that contain A.

Given any linear map $T \colon X \to Y$ of real vector spaces, and $C \subseteq X$ a convex subset of Y, then $T^{-1}(C)$ is convex in X. If X^* is the dual of any complex vector space, then since X is also a real vector space, it then follows that the sets

$$A_0 = \{x : |f(x)| \leq 1\}$$
$$A_1 = \{x : \text{Re}(f(x)) \leq 1\}$$

$$A_3 = \{x : \mathrm{Re}(f(x)) > 1\}$$

are all convex.

Proposition: Let X be a TVS, and let A be a convex subset of X. Then,

(a) $\overline{A}$ is convex;

(b) if $a \in A^\circ$ and $b \in \overline{A}$, then

$$[a,b) := \{tb + (1-t)a : 0 \le t < 1\}$$

is contained in A°.

Proof. Let $a \in A$, $b \in \overline{A}$, and let $0 \le t \le 1$. Let $(x_i)_i$ be a net in A such that $x_i \to b$. Then, $tx_i + (1-t)a \to tb + (1-t)a$. In particular, this means that if $b \in \overline{A}$ and $a \in A$, then $[a,b] \subseteq \overline{A}$. So, given any $a, b \in \overline{A}$, we have that the net $(x_i)_i \to b$ is such that $[x_i, a] \subseteq \overline{A}$, so it follows that $[b,a] \subseteq \overline{A}$, whence $\overline{A}$ is convex.

For (b), fix $0 < t < 1$, and set $c = tb + (1-t)a$ for some $a \in A^\circ$ and $b \in \overline{A}$. Since $a \in A^\circ$, there is an open neighborhood V of 0 such that $a + V \subseteq A$. Therefore, for any $d \in A$, we have

$$\begin{aligned} A &\supseteq td + (1-t)(a+V) \\ &= t(d-b) + tb + (1-t)(a+V) \\ &= (t(d-b) + (1-t)V) + c. \end{aligned}$$

We hope to find some $d \in A$ such that $0 \in t(d-b) + (1-t)V$, which we call U. The preceding inclusion then shows that $c \in A^\circ$, since U is an open neighborhood of 0 with $c + U \subseteq A$.

Finding such a d is equivalent to finding d such that $0 \in \frac{1}{t}(1-t)V + (d-b)$, or $d \in b - \frac{1}{t}(1-t)V$,. Yet, we know that $0 \in -\frac{1}{t}(1-t)V$ by the assumption that V is an open neighborhood of 0. Since $b \in \overline{A}$, it follows that $b - \frac{1}{t}(1-t)V \cap A \neq \emptyset$ by the definition of closure, whence such a $d \in A$ exists. □

Corollary: If $A \subseteq X$, then $\overline{\mathrm{conv}}(A) = \overline{\mathrm{conv}(A)}$.

Definition: A set $A \subseteq X$ is called balanced if for any $|\alpha| \le 1$ and any $x \in A$, we have $\alpha x \in A$.

We say a set A is *absorbing* if for any $x \in X$, there is some $\varepsilon > 0$ such that $tx \in A$ for all $0 \le t < \varepsilon$. We say A is absorbing at a if for any $x \in X$, there is $\varepsilon > 0$ such that $a + tx \in A$ for all $0 \le t < \varepsilon$.

Given a seminorm p on V, the set $V = \{x : p(x) < 1\}$ is a convex balanced set that is absorbing at each of its points. We will show that the converse actually holds — that is, a vector space is a LCS if and only if the set of all open, convex, balanced subsets of X forms a neighborhood basis at 0.

Proposition: Suppose X is a vector space over F, and V is a nonempty convex balanced set that is absorbing at each of its points. Then, there is a unique seminorm p on X such that

$$V = \{x \in X : p(x) < 1\}.$$

Proof. Let

$$p(x) = \inf\{t \ge 0 : x \in tV\}.$$

Since V is absorbing, it follows that $X = \bigcup_{n=1}^{\infty} nV$, so this is a well-defined quantity. Furthermore, $p(0) = 0$.

To show that $p(\alpha x) = |\alpha| p(x)$, we suppose $\alpha \neq 0$, and since V is balanced,

$$\begin{aligned} p(\alpha x) &= \inf\{t \ge 0 : \alpha x \in tV\} \\ &= \inf\left\{t \ge 0 : x \in \frac{t}{\alpha} V\right\} \\ &= \inf\left\{t \ge 0 : x \in \frac{t}{|\alpha|} V\right\} \end{aligned}$$

$$= |\alpha| \inf\left\{ \frac{t}{|\alpha|} : x \in \frac{t}{|\alpha|} V \right\}$$
$$= |\alpha| p(x).$$

We now observe that if $\alpha, \beta \geq 0$, then for any $a, b \in V$, we have

$$\alpha a + \beta b = (\alpha + \beta)\left(\frac{\alpha}{\alpha+\beta} a + \frac{\beta}{\alpha+\beta} b \right)$$
$$\in (\alpha + \beta) V,$$

since V is convex. For any $x, y \in X$, take $p(x) = \alpha$ and $p(y) = \beta$, and let $\delta > 0$. Then, $x \in (\alpha + \delta)V$ and $y \in (\beta + \delta)V$ by the definition of the infimum, so that $x + y \in (\alpha + \beta + 2\delta)V$. Therefore, we have $p(x+y) \leq \alpha + \beta + 2\delta$, which holds for all $\delta > 0$, so that $p(x+y) \leq p(x) + p(y)$.

Finally, we will have to show that $V = \{x : p(x) < 1\}$. If $p(x) = \alpha < 1$, then for any $\alpha < \beta < 1$, we have $x \in \beta V \subseteq V$, which emerges from the fact that V is balanced. Thus, $V \supseteq \{x : p(x) < 1\}$. If $x \in V$, then $p(x) \leq 1$, and since V is absorbing at x, there is some $\varepsilon > 0$ such that for $0 < t < \varepsilon$, we have $y = x + tx \in V$. Yet, this means $x = \frac{1}{1+t} y$ with $y \in V$, so that $p(x) = \frac{1}{1+t} p(y) \leq \frac{1}{1+t} < 1$. □

The seminorm p is called the Minkowski function for V, or the *gauge* of V.

Proposition: Let X be a TVS, and let $\mathcal{U}$ be the collection of open, convex, balanced subsets of X. Then, X is locally convex if and only if $\mathcal{U}$ is a basis for the neighborhood system at 0.

Proof. Suppose X is locally convex with topology defined by the family of seminorms $\mathcal{P}$. Then, we observe that each of the subbasic open neighborhoods of 0,

$$\{U(0, p, \varepsilon) : \varepsilon > 0, p \in \mathcal{P}\},$$

are open, convex, and balanced; since the set of all finite intersections of all these subbasic open neighborhoods yields a neighborhood basis for 0, and finite intersections of open, balanced, convex subsets are open, balanced, and convex, so it follows that $\mathcal{U}$ contains the set of all finite intersections of all these subbasic open neighborhoods, whence $\mathcal{U}$ contains a basis for the neighborhood system at 0. In particular, this means $\mathcal{U}$ also must generate the neighborhood system at 0, so $\mathcal{U}$ forms a basis for the neighborhood system at 0.

Now, suppose $\mathcal{U}$ forms a basis for the neighborhood system at 0. Observe that an open, convex, balanced subset of V contains 0, and is itself absorbing. We claim that the family of Minkowski functionals

$$\mathcal{P} = \{p_V(\cdot - z) : z \in X, V \in \mathcal{U}\}$$

generates the topology for X. □

Thus, it follows that in a locally convex space, a net $(x_i)_i$ converges in the topology to x if and only if for all seminorms $p \in \mathcal{P}$, we have $p(x_i - x) \to 0$.

Example:

(i) Suppose X is a locally compact Hausdorff space (or any Tychonoff space, but we will work with these for uniformity), and let $C(X)$ be the space of all continuous functions on X. Then, the family of seminorms $p_K(f) = \sup_{x \in K} |f(x)|$ yields a locally convex space where convergence is defined as uniform convergence on compact subsets.

(ii) If X is a normed space, then the family of seminorms $p_\varphi(x) = \left|\varphi(x)\right|$ for each $\varphi \in X^*$ defines the weak topology on X.

(iii) If X is a normed space, then the family of seminorms $p_x(\varphi) = \left|\varphi(x)\right|$ for each $x \in X$ defines the weak*, or w^*, topology on X^*.

The weak and weak* topologies are very important, and we will discuss them more in depth.

Metrizability and Normability

When we look at a locally convex space, we concern ourselves with whether the topology of the space is given by a metric or not. It turns out this is (effectively) equivalent to there being countably many seminorms defining the topology.

Proposition: Let $\{p_n : n \geq 1\}$ be a sequence of separating seminorms on X. For any $x, y \in X$, we may take

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \frac{p_n(x-y)}{1+p_n(x-y)}.$$

Then, the topology on X is defined by this metric.

Definition: A Fréchet space is a locally convex space whose topology is defined by a metric that is complete.

Definition: If X is a topological vector space, and $B \subseteq X$, then we say B is bounded if, for every open set U containing 0, there is some $\varepsilon > 0$ such that $\varepsilon B \subseteq U$.

Proposition: If X is a locally convex space, then X is normable if and only if X has a nonempty bounded open set.

Proof. The open unit ball U_X is bounded. Therefore, we only need to show one direction. Suppose X is a locally convex space with a bounded open set U. Translating if necessary, we may assume $0 \in U$.

Since X is locally convex, there is an open, convex, balanced subset of U which we call V; using the gauge of V, we may find a continuous seminorm p such that $\{x : p(x) < 1\} = V$. We will show that p is a norm and that p defines the topology on X.

First, let $0 \neq x \in X$. Let W_0 and W_x be disjoint open subsets with $0 \in W_0$ and $x \in W_x$. Since W_0 is absorbing, there is some $\varepsilon > 0$ such that $W_0 \supseteq \varepsilon U \supseteq \varepsilon V$. We note that $\varepsilon V = \{y : p(y) < \varepsilon\}$. Since $x \notin W_0$, we have $p(x) \geq \varepsilon$, so that p is a norm.

To show that the topology on X is generated by p, it suffices to show that if q is any continuous seminorm, then there is some α such that $q \leq \alpha p$. Since q is continuous, it follows that there is some $\varepsilon > 0$ such that $\{x : q(x) < 1\} \supseteq \varepsilon U \supseteq \varepsilon V$. Yet, this means that if $p(x) < \varepsilon$, then $q(x) < 1$, whence $q \leq \varepsilon^{-1} p$. □

Hyperplane Separation and Continuous Linear Functionals

We recall the classification of continuous linear functionals in a locally convex topological vector space.

Theorem: If X is a locally convex topological vector space with defining seminorms $\mathcal{P}$, and $f : X \to \mathbb{C}$ is a linear functional, then the following statements are equivalent:

(i) f is continuous;

(ii) f is continuous at 0;

(iii) f is continuous at some point;

(iv) $\ker(f)$ is closed;

(v) there are $p_1, \ldots, p_n \in \mathcal{P}$ and some constant C such that $|f(x)| \leq C \max\{p_1(x), \ldots, p_n(x)\}$.

To understand hyperplane separation, we start with the most basic case — namely, that of separating a convex subset from the origin via a continuous linear functional.

Theorem: If X is a topological vector space, and G is an open convex nonempty subset of X that does not contain the origin, then there is a closed hyperplane M such that $M \cap G = \emptyset$.

Proof. We start by assuming that X is an $\mathbb{R}$-linear space. Choose $x_0 \in G$, and let $H = x_0 - G$. Then, H is an open, convex subset of X that contains 0. Taking the gauge of H yields a continuous sublinear functional $q : X \to \mathbb{R}$ with $H = \{x : q(x) < 1\}$. Furthermore, since $x_0 \notin H$, we have that $q(x_0) \geq 1$.

Consider the subspace $Y = \{\alpha x_0 : \alpha \in \mathbb{R}\}$. Let $f_0 : Y \to \mathbb{R}$ be defined by taking $f_0(\alpha x_0) = \alpha q(x_0)$. Observe that if $\alpha \geq 0$, then $f_0(\alpha x_0) = q(\alpha x_0)$, and if $\alpha < 0$, then $f_0(\alpha x_0) = \alpha q(x_0) \leq \alpha < 0 \leq q(\alpha x_0)$, whence $f_0 \leq q$ on

Y. Using the Hahn–Banach–Minkowski extension yields some $f: X \to \mathbb{R}$ a continuous linear functional such that $f|_Y = f_0$ and $f \leq q$ on X. Then, $M = \ker(f)$ is the desired hyperplane.

If $x \in G$, then $x_0 - x \in H$, so $f(x_0) - f(x) = f(x_0 - x) \leq q(x_0 - x) < 1$, while $f(x) > f(x_0) - 1 = q(x_0) - 1 \geq 0$ for all $x \in G$, whence $M \cap G = \emptyset$.

If X is a $\mathbb{C}$-linear space, then since X is also an $\mathbb{R}$-linear space, we obtain a continuous $\mathbb{R}$-linear functional $f: X \to \mathbb{R}$ with $G \cap \ker(f) = \emptyset$. If $F(x) = f(x) - if(ix)$, then F is a $\mathbb{C}$-linear functional with $f = \text{Re}(F)$ with $F(x) = 0$ if and only if $f(x) = f(ix) = 0$, so that $M = \ker(F)$ is the desired closed hyperplane. □

Affine hyperplanes in X are given by $M' = M + x_0$ for some $x_0 \in X$ and hyperplane M. It turns out that we can extend any separating affine (closed) subspace of a topological vector space to an affine hyperplane.

Corollary: Let X be a topological vector space, and let G be an open, convex, nonempty subset of X. If Y is an affine subspace of X with $Y \cap G = \emptyset$, then there is a closed affine hyperplane M in X such that $Y \subseteq M$ and $M \cap G = \emptyset$.

Proof. Choosing some $x_0 \in Y$, we may consider $G - x_0$ and $Y - x_0$, so that Y may be assumed to be a subspace of X. Let $Q: X \to X/Y$ be the quotient map. Notice that $Q^{-1}(Q(G)) = \{y + G : y \in Y\}$, so that $Q(G)$ is open in X/Y. Furthermore, $Q(G)$ is also convex since Q is a linear map. Furthermore, since $Y \cap G = \emptyset$, $0 \notin Q(G)$. In particular, there is then a closed hyperplane N in X/Y such that $N \cap Q(G) = \emptyset$. Setting $M = Q^{-1}(N)$, we have that M is our desired hyperplane. □

In the special case of a real topological vector space, it's useful to note that $X \setminus \ker(f)$ for a continuous linear functional $f: X \to \mathbb{R}$ has two connected components. This yields the following notion.

Definition: Let X be a real topological vector space. A subset $S \subseteq X$ is called an open half space if there is some continuous linear functional $f: X \to \mathbb{R}$ such that $S = f^{-1}((\alpha, \infty))$. We call S a closed half-space if there is a continuous linear functional $f: X \to \mathbb{R}$ such that $S = f^{-1}([\alpha, \infty))$.

We say A and B are strictly separated if they are contained in disjoint open half-spaces, while they are separated if they are contained in two closed half-spaces whose intersection is of the form $M = f^{-1}(\{\alpha\})$. That is, their intersection is a closed affine hyperplane. Notice also that continuous linear functionals are open maps, so that the interior of a closed half-space is an open half-space, and the closure of an open half-space is a closed half-space.

Theorem (General Hyperplane Separation)**:** Let X be a real topological vector space, and A and B disjoint convex subsets with A open. Then, there is a continuous linear functional $f: X \to \mathbb{R}$ and a real scalar α such that $A \subseteq f^{-1}((-\infty, \alpha))$ and $B \subseteq f^{-1}([\alpha, \infty))$.

Proof. Let $G = A - B$. Then, G is convex, and $G = \bigcup_{b \in B}(A - b)$, meaning G is open. Since $A \cap B = \emptyset$, it follows that $0 \notin G$. Thus, there is a closed hyperplane M in X such that $M \cap G = \emptyset$.

Let $f: X \to \mathbb{R}$ be a linear functional such that $M = \ker(f)$. Notice that $f(G)$ is a convex subset of $\mathbb{R}$ with $0 \notin f(G)$. Either, $f(x) > 0$ for all $x \in G$ or $f(x) < 0$ for all $x \in G$. Without loss of generality, we may assume that $f(x) > 0$ for all $x \in G$, so that $f(b) < f(a)$ for any $a \in A$ and $b \in B$, meaning there is some real number α such that $\sup_{b \in B} f(b) \leq \alpha \leq \inf_{a \in A} f(a)$. Since $f(A)$ and $f(B)$ are open intervals if B is open, it follows that $f < \alpha$ on B and $f > \alpha$ on A. □

Lemma: If X is a topological vector space, K is a compact subset of X, and V is an open subset satisfying $K \subseteq V$, then there is an open neighborhood U of 0 such that $K + U \subseteq V$.

Proof. Let $\mathcal{O}_0$ denote the system of open neighborhoods of 0. Suppose toward contradiction that for each $U \in \mathcal{O}_0$ that $K + U$ is not contained in V. We may choose vectors $x_U \in K$ and $y_U \in U$ such that $x_U + y_U \in V^c$. This yields a net $(y_U)_{U \in \mathcal{O}_0}$ and $(x_U)_{U \in \mathcal{O}_0}$ with $y_U \to 0$ and a subnet $x_{U'} \to x \in K$. Yet, this yields $(x_{U'} + y_{U'}) \to x$, implying that $x \in \overline{V^c} = V^c$, which yields a contradiction. □

Theorem (Hyperplane Separation for Locally Convex Spaces)**:** Let X be a real locally convex topological vector space, and let A and B be two disjoint closed convex subsets of X with B compact. Then, A and B are contained in disjoint open half-spaces.

Proof. Notice that B is a compact subset of A^c. Therefore, there is an open neighborhood U_1 of 0 such that $B + U_1 \subseteq A^c$. Since X is locally convex, we may find a continuous seminorm p on X such that $\{x : p(x) < 1\} \subseteq U_1$.

Define $U = \{x : p(x) < 1/2\}$. Then, $(B+U) \cap (A+U) = \emptyset$, and $A+U$ and $B+U$ are open convex subsets of X that contain A and B. Therefore, the result follows from the general hyperplane separation. □

Rephrasing this theorem for locally convex spaces by using the fact that any linear functional φ over $\mathbb{C}$ is given by $\varphi(x) = \text{Re}(\varphi(x)) - i\,\text{Re}(\varphi(ix))$ yields the following.

Theorem (Hyperplane Separation for Locally Convex Spaces, Encore)**:** Let X be a locally convex space, and let A and B be disjoint closed convex subsets of X with B compact. Then, there is a continuous linear functional $f : X \to \mathbb{C}$, some $\alpha \in \mathbb{R}$, and some $\varepsilon > 0$ such that for all $a \in A$ and $b \in B$, we have

$$\text{Re}(f(a)) \leq \alpha < \alpha + \varepsilon \leq \text{Re}(f(b)).$$

We will now discuss the dual space of one particular locally convex space.

Proposition: Let X be a locally compact Hausdorff space, and let $C(X)$ be topologized by uniform convergence on compact subsets. If $L : C(X) \to \mathbb{F}$ is a continuous linear functional, then there is a compact set K and a regular (complex) Borel measure μ on K such that $L(f) = \int_K f\,d\mu$ for any $f \in C(X)$. Conversely, all these measures define elements of $C(X)^*$.

Proof. Any compactly support measure defines an element of $C(X)^*$, and satisfies $|L(f)| \leq \|\mu\| \sup_{x\in K} |f(x)|$ for any such measure μ.

Conversely, let $L \in C(X)^*$. By the classification of continuous linear functionals, there are compact sets $K_1, \ldots, K_n$ and a constant C such that $|L(f)| \leq C \max\{p_{K_j}(f) : j = 1, \ldots, n\}$. Let $K = \bigcup_{j=1}^n K_j$. Then, $|L(f)| \leq \alpha p_K(f)$. In particular, for any $f \in C(X)$ with $f|_K = 0$, then $L(f) = 0$.

Define a linear functional $F : C(K) \to \mathbb{F}$ as follows. For any $g \in C(K)$, let $\widetilde{g}$ be a continuous extension to X, and set $F(g) = L(\widetilde{g})$. Suppose $\widetilde{g}_1$ and $\widetilde{g}_2$ are any such extensions; then, $g_1 - g_2 = 0$ on K, so that $L(\widetilde{g}_1) = L(\widetilde{g}_2)$, so that F is well-defined. Furthermore, F is linear. Since, for any $g \in C(K)$ and extension $\widetilde{g} \in C(X)$, we have $|F(g)| \leq C p_K(g) = C\|g\|$. In particular, this means F is continuous, so there is some regular complex Borel measure μ in $M(K)$ such that $F(g) = \int g\,d\mu$. Notice that if $f \in C(X)$ is such that $f|_K = g \in C(K)$, it follows that $L(f) = F(g) = \int f\,d\mu$. □

Weak Topologies and Extremal Structure

Duality

Let X be a locally convex space. Then, X^* denotes the space of continuous linear functionals on X, and admits a vector space structure. We will write $\langle x, \varphi\rangle = \varphi(x)$ whenever $x \in X$ and $\varphi \in X^*$. We will write $\hat{x} = \langle x, \cdot\rangle$ to denote the analogous linear functional on φ such that $\hat{x}(\varphi) = \langle x, \varphi\rangle$.

Definition: If X is a locally convex space, then the weak topology on X is the topology defined by the family of seminorms

$$p_\varphi(x) = \big|\langle x, \varphi\rangle\big|.$$

The weak* topology on X is the topology defined by the seminorms

$$p_x(\varphi) = \big|\langle x, \varphi\rangle\big|.$$

We have that $(x_i)_i \xrightarrow{\text{wk}} x$ if, for all $\varphi \in X^*$, $\varphi(x_i) \to \varphi(x)$. Similarly, we have that $(\varphi_i)_i \xrightarrow{\text{wk}^*} \varphi$ if, for all $x \in X$, $\varphi_i(x) \to \varphi(x)$.

Theorem: If X is a locally convex space, then $(X^*, \text{wk}^*)^* = X$.

Proof. First, if $x \in X$, then $\varphi \mapsto \langle x, \varphi\rangle$ is a weak*-continuous linear functional on X^* by definition.

Conversely, if $f \in (X^*, \mathrm{wk}^*)^*$, then there are vectors $x_1, \dots, x_n \in X$ such that

$$\left|f(\varphi)\right| \leq \sum_{k=1}^{n} \left|\langle x_k, \varphi\rangle\right|$$

for all $\varphi \in X^*$. This implies that

$$\bigcap_{k=1}^{n} \ker(\hat{x}_k) \subseteq \ker(f),$$

so there are $\alpha_1, \dots, \alpha_n$ such that $f = \sum_{k=1}^{n} \alpha_k \hat{x}_k$, meaning that $f \in X$. □

Local convexity is a very useful property to have, as the following result exhibits.

Theorem: Let X be a locally convex space, and let A be a convex subset of X. Then, $\overline{A} = \overline{A}^{\mathrm{wk}}$.

Proof. Notice that the weak topology is weaker than the topology on X, so $\overline{A} \subseteq \overline{A}^{\mathrm{wk}}$.

Conversely, suppose there were $x \in \overline{A}^{\mathrm{wk}} \setminus \overline{A}$. Then, by geometric Hahn–Banach, there is $\varphi \in X^*$, $\alpha \in \mathbb{R}$, and $\varepsilon > 0$ such that

$$\mathrm{Re}(\langle a, \varphi\rangle) \leq \alpha < \alpha + \varepsilon \leq \mathrm{Re}(\langle x, \varphi\rangle)$$

for all $a \in \overline{A}$. Yet, this means that $\overline{A}^{\mathrm{wk}} \subseteq \{y \in X : \mathrm{Re}(\langle y, \varphi\rangle) \leq \alpha\} =: B$, even though the latter is weakly closed. Since $x \notin B$, it follows that $x \notin \overline{A}^{\mathrm{wk}}$, which yields a contradiction. □

Corollary: A convex subset of X is closed if and only if it is weakly closed.

Duality in Subspaces and Quotient Spaces

Proposition: Let p be a seminorm on X, M a subspace of X, and $\overline{p} \colon X/M \to [0, \infty)$ is defined by

$$\overline{p}(x + M) = \inf\{p(x + y) : y \in M\},$$

then $\overline{p}$ is a seminorm on X/M. If $\mathcal{P}$ is a family of continuous seminorms on X defining a locally convex topology, then $\overline{\mathcal{P}}$ defines the quotient topology on X/M.

In particular, if X is a locally convex space and $M \leq X$ is a subspace, then X/M is a locally convex space. Suppose $f \in (X/M)^*$. Then, $f \circ \pi \in X^*$, where $\pi \colon X \to X/M$ is the projection. Furthermore, observe that $f \circ \pi$ an element of

$$M^{\perp} = \{\varphi \in X^* : \varphi(m) = 0 \text{ for all } m \in M\}.$$

In particular, we obtain the map $f \mapsto f \circ \pi$, yielding a linear map $(X/M)^* \to M^{\perp} \subseteq X^*$.

Theorem: Let X be a locally convex space, $M \leq X$ a subspace, and $\pi \colon X \to X/M$ the natural map. Then, $f \mapsto f \circ \pi$ defines a bijection between $(X/M)^*$ and $M^{\perp}$.

Equipping $(X/M)^*$ with the weak* topology, and $M^{\perp}$ with the relative weak* topology, yields that this bijection is a homeomorphism. If X is a normed space, then the bijection is an isometry and a weak*–weak* homeomorphism.

Proof. Define $\rho \colon (X/M)^* \to M^{\perp}$ by $\rho(f) = f \circ \pi$. Then, we have that ρ is well-defined and takes $(X/M)^*$ to $M^{\perp}$. Notice that if $0\rho(f)$, then $f \circ \pi = 0$, but since π is surjective, f must be the zero map, whence ρ is injective.

Let $\varphi \in M^{\perp}$, and let $f \colon X/M \to \mathbb{C}$ be given by $f(x + M) = \langle x, \varphi\rangle$. Since $M \subseteq \ker(\varphi)$, f is well-defined and linear. Since

$$\pi^{-1}(\{x + M : |f(x + M)| < 1\}) = \left\{x \in X : \left|\langle x, \varphi\rangle\right| < 1\right\}$$

is open in X, it follows that $\{x+M : |f(x+M)| < 1\}$ is open in X/M, so f is continuous.

If X is a normed space, then results from Banach space theory yields that ρ is an isometry. If $(f_i)_i \subseteq (X/M)^*$ converges to 0 in the weak* topology on $(X/M)^*$, then for each $x \in X$, we have that

$$\begin{aligned}\langle x, \rho(f_i)\rangle &= f_i(\pi(x)) \\ &\to 0,\end{aligned}$$

so that $\rho(f_i) \to 0$ weak* in X^*. Conversely, if $\rho(f_i) \to 0$ weak* in X^*, it follows that for each $x \in X$, we have $f_i(x+M) = \langle x, \rho(f_i)\rangle \to 0$ weak* in $(X/M)^*$. □

Banach–Alaoglu Theorem and Applications

Theorem: If X is a normed space, then the unit ball B_{X^*} of X^* is weak*-compact.

Proof. Let $D_x = \{\alpha \in \mathbb{C} : |\alpha| \leq 1\}$. Set $D = \prod_{x \in B_X} D_x$. By Tychonoff's theorem, D is compact. Let $\tau\colon B_{X^*} \to D$ be defined by

$$\tau(\varphi)(x) = \langle x, \varphi\rangle.$$

Then, $\tau(\varphi) \in D$ with x coordinate given by $\langle x, \varphi\rangle$. We will show that τ is a homeomorphism from B_{X^*} with the weak* topology onto $\tau(B_{X^*}) \subseteq D$ with the subspace topology, and that $\tau(B_{X^*}) \subseteq D$ is closed.

Suppose $\tau(\varphi_1) = \tau(\varphi_2)$. Then, for each $x \in B_X$, $\langle x, \varphi_1\rangle = \langle x, \varphi_2\rangle$. Since B_X separates the points of X^*, it follows that $\varphi_1 = \varphi_2$.

Let $(\varphi_i)_i$ be a net in B_{X^*} with $\varphi_i \xrightarrow{\text{wk}^*} \varphi$. For each $x \in B_X$, it follows that $\tau(\varphi_i)(x) \to \tau(\varphi)(x)$, or that each coordinate of $(\tau(\varphi_i))_i$ converges to $\tau(\varphi)$, so that $\tau(\varphi_i) \to \tau(\varphi)$, meaning τ is continuous.

Let $(\varphi_i)_i$ be a net in B_{X^*}, $f \in D$, and $\tau(\varphi_i) \to f$ in D. Then, $f(x) = \lim\langle x, \varphi_i\rangle$ exists for each $x \in B_X$. If $x \in X$, we choose $\alpha > 0$ such that $\|\alpha x\| \leq 1$. Take $f(x) = \alpha^{-1} f(\alpha x)$.

If $\beta > 0$ is also such that $\|\beta x\| \leq 1$, then

$$\begin{aligned}\alpha^{-1} f(\alpha x) &= \alpha^{-1} \lim\langle \alpha x, \varphi_i\rangle \\ &= \beta^{-1} \lim\langle \beta x, \varphi_i\rangle \\ &= \beta^{-1} f(\beta x),\end{aligned}$$

so that $f(x)$ is well-defined. Notice that by the way f is defined, it is necessarily a linear functional. If $\|x\| \leq 1$, then $f(x) \in D_x$, so $|f(x)| \leq 1$, meaning that there is $\varphi \in B_{X^*}$ such that $\tau(\varphi) = f$. Thus, $\tau(B_{X^*})$ is closed in D, so $\tau(B_{X^*})$ is compact. □

Recall that a Banach space X is called reflexive if the embedding of X into its double dual X^{**} is surjective, where we recall that $\hat{x} \in X^{**}$ is defined by

$$\langle \varphi, \hat{x}\rangle = \langle x, \varphi\rangle.$$

We write $\iota\colon X \mapsto X^{**}$, taking $x \mapsto \hat{x}$, for this canonical embedding.

We will usually use σ-notation for weak topologies. That is, if X is a locally convex space, and X^* is the continuous dual space, then

- $\sigma(X, X^*)$ denotes the weak topology on X induced by X^*;
- $\sigma(X^*, X)$ denotes the weak* topology on X^* induced by X.

Recall that convergence of nets in these topologies is given by

- $(x_i)_i \xrightarrow{\sigma(X,X^*)} x$ if and only if for all $\varphi \in X^*$, $\langle x_i, \varphi\rangle \to \langle x, \varphi\rangle$;
- $(\varphi_i)_i \xrightarrow{\sigma(X^*,X)} \varphi$ if and only if for all $x \in X$, $\langle x, \varphi_i\rangle \to \langle x, \varphi\rangle$ if and only if for all $x \in X$, $\langle \varphi_i, \hat{x}\rangle \to \langle \varphi, x\rangle$.

Theorem (Goldstine's Theorem)**:** Let X be a normed space. Then, $\iota(B_X)$ is $\sigma(X^{**}, X^*)$-dense in X^{**}.

Proof. We will denote by w^{**} the weak* topology on X^{**} induced by X^*. Let $B = \overline{B_X}^{w^{**}}$, and notice that $B \subseteq B_{X^{**}}$. Suppose there is some $\theta \in B_{X^{**}} \setminus B$. The Hahn–Banach theorem implies there is some $\varphi \in X^*$, $\alpha \in \mathbb{R}$, and $\varepsilon > 0$ such that

$$\operatorname{Re}(\langle x, \varphi \rangle) < \alpha < \alpha + \varepsilon < \operatorname{Re}(\langle \varphi, \theta \rangle)$$

for all $x \in B_X$. Since $0 \in B_X$, it follows that $0 < \alpha$. Dividing, we may assume that there is $\varphi \in X^*$ and $\varepsilon > 0$ such that

$$\operatorname{Re}(\langle x, \varphi \rangle) < 1 < 1 + \varepsilon < \operatorname{Re}(\langle \varphi, \theta \rangle).$$

Notice that $\left|\langle x, \varphi \rangle\right| \leq 1$ (multiplying by a phase) for any $\|x\| \leq 1$, so that $\varphi \in B_{X^*}$. Yet, we also have

$$\begin{aligned} 1 + \varepsilon &< \operatorname{Re}(\langle \varphi, \theta \rangle) \\ &\leq \left|\langle \varphi, \theta \rangle\right| \\ &\leq \|\theta\| \\ &\leq 1, \end{aligned}$$

which is a contradiction. □

Theorem: Let X be a Banach space. The following are equivalent:

(a) X is reflexive;

(b) X^* is reflexive;

(c) we have $\sigma(X^*, X) = \sigma(X^*, X^{**})$;

(d) B_X is weakly compact.

Proof. If X is reflexive, then we have that $(\varphi_i)_i \to \varphi$ weak* if and only if $\langle x, \varphi_i \rangle \to \langle x, \varphi \rangle$ for all $x \in X$ if and only if $\langle \varphi_i, \hat{x} \rangle \to \langle \varphi, \hat{x} \rangle$ for all $x \in X$ if and only if $\langle \varphi_i, \theta \rangle \to \langle \varphi, \theta \rangle$ for all $\theta \in X^{**}$. This gives (a) implies (c).

If the unit ball B_X of X is weakly compact, then it is the case that $\iota(B_X) \subseteq B_{X^{**}}$ is $\sigma(X^{**}, X^*)$-closed. Yet, Goldstine's theorem gives that $\iota(B_X)$ is dense in $B_{X^{**}}$, so that $\iota(B_X) = B_{X^{**}}$, whence $\iota(X) = X^{**}$. This gives (d) implies (a).

By the Banach–Alaoglu theorem, B_{X^*} is $\sigma(X^*, X)$-compact. By the assumption in (c), we have $\sigma(X^*, X^{**}) = \sigma(X^*, X)$, so B_{X^*} is $\sigma(X^*, X^{**})$-compact. Therefore, since (d) implies (a), it follows that X^* is reflexive, giving (b).

We assume (b). The unit ball B_X is norm-closed in X^{**}. Therefore, since B_X is convex, it follows that B_X is $\sigma(X^{**}, X^{***})$-closed. Since $X^* \cong X^{***}$, it follows that $\iota(B_X)$ is $\sigma(X^{**}, X^*)$-closed. By Goldstine's theorem, we know that B_X is $\sigma(X^{**}, X^*)$-dense in $B_{X^{**}}$, so that $B_X = B_{X^{**}}$, whence X is reflexive. This yields (a).

Finally, assuming (a), we have that $B_{X^{**}}$ is $\sigma(X^{**}, X^*)$-compact in $B_{X^{**}}$. Since $X \cong X^{**}$, it follows that B_X is $\sigma(X, X^*)$-compact. □

Corollary: If X is a reflexive Banach space, and $M \leq X$ is a closed subspace, then M is a reflexive Banach space.

Proof. Notice that the unit ball of M is the intersection of M with the unit ball of X, so the unit ball of M is $\sigma(X, X^*)$-compact. Since $\sigma(X, X^*)|_M = \sigma(M, M^*)$, it follows that the unit ball of M is weakly compact with respect to $\sigma(M, M^*)$. □

Corollary: If X is reflexive, then X is weakly sequentially complete.

Proof. Let $(x_n)_n$ be weakly Cauchy. Then, $(\langle x_n, \varphi \rangle)_n$ is Cauchy for each $\varphi \in X^*$. So, by the principle of uniform boundedness, $\|x_n\| \leq M$ for some constant M.

Since X is reflexive, $\{x \in X : \|x\| \leq M\}$ is weakly compact. So, there is some $x \in X$ such that $\left(x_{n_k}\right)_{n_k} \to x$ weakly for some subsequence of $(x_n)_n$. Since $(\langle x_n, \varphi\rangle)_n$ converges for all $\varphi \in X^*$, it follows that $\langle x_n, \varphi\rangle \to \langle x, \varphi\rangle$ for all $\varphi \in X^*$, whence $(x_n)_n \to x$ weakly. □

Corollary: If X is a reflexive Banach space, $M \leq X$ is a closed subspace, and $x_0 \in X \setminus M$, then there is some $y_0 \in M$ such that $\left\|x_0 - y_0\right\| = \text{dist}_M(x_0)$.

Proof. The map $x \mapsto \|x - x_0\|$ is weakly lower semicontinuous. If $d = \text{dist}_M(x_0)$, then the set $M \cap \{x : \|x - x_0\| \leq 2d\}$ is weakly compact. Since lower semicontinuous functions attain their minima on compact sets, the result follows. □

Closed subspaces satisfying the property that for any $x_0 \in X \setminus M$, there exists $y_0 \in M$ such that $\left\|x_0 - y_0\right\| = \text{dist}_M(x_0)$ are known as proximinal subspaces. Not every subspace of a normed vector space is proximinal.

Proposition: Let X be a Banach space, $\varphi \in X^*$. Then, $\ker(\varphi)$ is proximinal if and only if there is some $x \in X$ with $\|x\| = 1$ such that $\langle x, \varphi\rangle = \left\|\varphi\right\|$.

Proof. Suppose $M = \ker(\varphi)$ is proximinal. If $f : X/M \to F$ is given by $f(x + M) = \langle x, \varphi\rangle$, then f is a linear functional with $\|f\| = \left\|\varphi\right\|$. Since X/M is one-dimensional, there is $x \in X$ such that $\|x + M\| = 1$ and $f(x + M) = \|f\|$. Since M is proximinal, there is $y \in M$ such that $1 = \|x + M\| = \left\|x + y\right\|$. Therefore,

$$\begin{aligned}\langle x + y, \varphi\rangle &= \langle x, \varphi\rangle \\ &= f(x + M) \\ &= \|f\| \\ &= \left\|\varphi\right\|.\end{aligned}$$

Now, suppose there is $x_0 \in X$ with $\|x_0\| = 1$ and $\langle x_0, \varphi\rangle = \left\|\varphi\right\|$. If $x \in X$ with $\|x + M\| = \alpha > 0$, then $\left\|\alpha^{-1}x + M\right\| = 1$; since $\|x_0 + M\| = 1$ and X/M is one-dimensional, it follows that there is $\beta \in \mathbb{C}$ with $\left|\beta\right| = 1$ such that $\alpha^{-1}x + M = \beta x_0 + M$, whence $\alpha^{-1}x - \beta x_0 \in M$. Equivalently, we have $x - \alpha\beta x_0 \in M$. Since

$$\begin{aligned}\left\|x - (x - \alpha\beta x_0)\right\| &= \left\|\alpha\beta x_0\right\| \\ &= \alpha \\ &= \text{dist}_M(x),\end{aligned}$$

it follows that the distance from x to M is attained at the element $x - \alpha\beta x_0 \in M$. □

Separability and Metrizability

Theorem: If X is a Banach space, then B_{X^*} with the weak* topology is metrizable if and only if X is separable.

Proof. Let X be separable, $(x_n)_n$ a countable dense subset of B_X. Let $D = \prod_{n=1}^{\infty} B_{\mathbb{C}}(0, 1)$. Since D is a compact metric space, it follows that if B_{X^*} with the weak* topology is homeomorphic to a subset of D, then B_{X^*} is weak* metrizable.

Let $\tau\colon B_{X^*} \to X$ be defined by $\tau(\varphi) = (\langle x_n, \varphi\rangle)_n \in D$. If $(\varphi_i)_i$ is a net in B_{X^*} converging weak* to $\varphi \in X^*$, it follows that for each $n \geq 1$, we have that $\tau(\varphi_i) \to \tau(\varphi)$, so τ is weak* continuous. Furthermore, $\tau(\varphi) = 0$ if and only if $\varphi(x_n) = 0$ for all n; since φ is then zero on a dense subset, it follows that φ is the zero functional, so τ is injective. Since B_{X^*} is compact, τ is a homeomorphism onto its image, so B_{X^*} is weak* metrizable.

Suppose B_{X^*} is weak* metrizable. Let $\{U_n : n \geq 1\}$ be a family of weak*-open neighborhoods of 0 whose intersection is 0. Then, for each n, there is some finite $F_n \subseteq X$ such that

$$\left\{\varphi \in B_{X^*} : \left|\langle x, \varphi\rangle\right| < 1 \text{ for all } x \in F_n\right\} \subseteq U_n.$$

Let $F = \bigcup_{n=1}^{\infty} F_n$. Then, F is countable. The pre-annihilator of $F^{\perp}$ is the closed linear span of F, which is

separable. Yet, if $\varphi \in F^\perp$, then for each $n \geq 1$ and each $x \in F_n$, we have

$$\begin{aligned}\left|\left\langle x, \frac{\varphi}{\|\varphi\|}\right\rangle\right| &= 0 \\ &< 1,\end{aligned}$$

so $\frac{\varphi}{\|\varphi\|} \in U_n$ for all $n \geq 1$, whence $\varphi = 0$. Therefore, $F^\perp = \{0\}$, meaning its pre-annihilator is X. Thus, X is separable. □

We may be curious about an analogous result for the weak topology. That is, if X^* is separable, is the weak topology on B_X metrizable? This follows immediately from the above theorem when we consider the embedding of X into X^{**}. However, this isn't very useful unless we're considering a separable reflexive space.

However, in general, if X is separable, it is not necessarily the case that B_X equipped with the weak topology is metrizable.

Theorem: If a sequence in ℓ^1 converges weakly, then the sequence converges in norm.

Proof. Since ℓ^1 is separable, we have that B_{ℓ^∞} is weak* metrizable. Since B_{ℓ^∞} is weak* compact (i.e., it is a compact metrizable space), we may use the Baire Category Theorem.

Let $(f_n)_n \to 0$ weakly in ℓ^1, and let $\varepsilon > 0$. For each $m \geq 0$, let

$$F_m = \left\{\phi \in B_{\ell^\infty} : \left|\langle f_n, \phi\rangle\right| \leq \varepsilon/3 \text{ for all } n \geq m\right\}.$$

Then, F_m is weak*-closed in B_{ℓ^∞}. Since $(f_n)_n \to 0$ weakly, it follows that the union of all such F_m is equal to B_{ℓ^∞}.

By the Baire Category Theorem, there is some F_m with nonempty interior. A metric on B_{ℓ^∞} inducing the weak* topology is given by

$$d(\phi, \psi) = \sum_{j=1}^{\infty} 2^{-j}\left|\phi(j) - \psi(j)\right|.$$

Since F_m has nonempty weak* interior, there is $\phi \in F_m$ and $\delta > 0$ such that $U(\phi, \delta) \subseteq F_m$, where we recall that

$$U(\phi, \delta) = \{\psi \in B_{\ell^\infty} : d(\phi, \psi) < \delta\}.$$

Let J be such that $2^{-(J-1)} < \delta$. Let $n \geq m$, and let $\psi \in \ell^\infty$ be defined by $\psi(j) = \phi(j)$ for each $1 \leq j \leq J$ and $\psi(j) = \operatorname{sgn}(f_n(j))$ for each $j > j$. Then, $\psi(j) f_n(j) = |f_n(j)|$ for each $j > J$. We have that $\psi \in B_{\ell^\infty}$.

Furthermore, we have

$$\begin{aligned}d(\phi, \psi) &= \sum_{j=J+1}^{\infty} 2^{-j}\left|\phi(j) - \psi(j)\right| \\ &\leq 2^{-(J-1)} \\ &< \delta.\end{aligned}$$

Thus, $\psi \in F_m$, so $\left|\langle f_n, \psi\rangle\right| \leq \varepsilon/3$ for all $n \geq m$. In particular, we have

$$\left|\sum_{j=1}^{J} \phi(j) f_n(j) + \sum_{j=J+1}^{\infty} |f_n(j)|\right| \leq \frac{\varepsilon}{3}$$

for all $n \geq m$.

Yet, since $(f_n)_n \to 0$ in ℓ^1, we have some $m_1 \geq m$ such that for all $n \geq m_1$, we have

$$\sum_{j=1}^{J} |f_n(j)| < \varepsilon/3.$$

Therefore, we have

$$\begin{aligned} \|f_n\| &= \sum_{j=1}^{\infty} |f_n(j)| \\ &< \frac{\varepsilon}{3} + \left| \sum_{j=J+1}^{\infty} |f_n(j)| + \sum_{j=1}^{J} \phi(j) f_n(j) \right| + \left| \sum_{j=1}^{J} \phi(j) f_n(j) \right| \\ &< \frac{2\varepsilon}{3} + \sum_{j=1}^{J} |f_n(j)| \\ &< \varepsilon \end{aligned}$$

whenever $n \geq m_1$. This gives norm convergence. □

In particular, if it were the case that B_{ℓ^1} with the weak topology was metrizable, then we would have that the weak and norm topologies on ℓ^1 agreed. Yet, this is not the case.

Extremal Structure

Definition: Let X be a vector space, and let $K \subseteq X$ be a convex set. We call a subset $F \subseteq K$ a *face* if F is a convex subset satisfying the following criterion: if $0 < t < 1$, and $x, y \in K$ with $(1-t)x + ty \in F$ then $x, y \in F$.

A singleton set $\{z\} \in K$ that is a face is called an extreme point. We write $\text{ext}(K)$ for the set of extreme points of K.

In particular, $z \in K$ is an extreme point if and only if, for any $0 < t < 1$, if there are $x, y \in K$ with $(1-t)x + ty = z$, then $x = y = z$. In fact, we can take $t = 1/2$ — that is, we only need to concern ourselves with the midpoint of any such line segment. To see this, we assume without loss of generality that $t < 1/2$. Then, the vector $w = (1-2t)x + 2ty$ lies in C since C is convex, and satisfies $\frac{1}{2}(x+w) = z$. In particular, this means that $x = z = w$, so that $y = z$.

There are convex subsets not admitting extreme points.

Example: Let $X = c_0$. We will show that the unit ball B_{c_0} has no extreme points.

Let $z \in \text{ext}\left(B_{c_0}\right)$. Notice that for any $0 < \|z\| < 1$, we have

$$z = (1 - \|z\|)0 + \|z\| \frac{z}{\|z\|},$$

meaning in particular that z cannot be an extreme point if $\|z\| < 1$. In particular, this means that $\|z\| = 1$.

Since $z \in c_0$, there is $n_0 \in \mathbb{N}$ with $|z(n_0)| < 1$. Write $z(n_0) = (1-t)\lambda + t\mu$ for some $0 < t < 1$ and $|\lambda| = \left|\mu\right| = 1$. Consider the sequences defined by

$$x(n) = \begin{cases} z(n) & n \neq n_0 \\ \lambda & n = n_0 \end{cases}$$

$$y(n) = \begin{cases} z(n) & n \neq n_0 \\ \mu & n = n_0. \end{cases}$$

Then, $\|x\|, \left\|y\right\| \leq 1$, but $(1-t)x + ty = z$ with $x \neq z$ and $y \neq z$, meaning that z cannot be extreme.

Proposition: Let Ω be a locally compact Hausdorff space.

(i) If Ω is not compact, then $\operatorname{ext}\left(B_{C_0(\Omega)}\right) = \emptyset$.

(ii) If Ω is compact, then

$$\operatorname{ext}\left(B_{C(\Omega)}\right) = \{u \in C(\Omega) : |u(x)| = 1 \text{ for all } x \in \Omega\}.$$

Proof. Let $f \in \operatorname{ext}\left(B_{C_0(\Omega)}\right)$. Then, $\|f\| = 1$. Suppose there is some x_0 with $|f(x_0)| < 1$. Choose δ such that $|f(x_0)| < \delta < 1$. By continuity, there is some open neighborhood U of x_0 with $|f| < \delta$ on U. By Urysohn's Lemma, there is a continuous function $g \colon \Omega \to [0,1]$ with $\operatorname{supp}(g) \subseteq U$ and $g(x_0) = 1$.

Then, we may write

$$f = \frac{1}{2}((f + (1-\delta)g) + (f - (1-\delta)g)).$$

First, we observe that on U^c, we have that $|f \pm (1-\delta)g| = |f| \leq 1$, while on U we have

$$\begin{aligned} |f \pm (1-\delta)g| &\leq |f| + (1-\delta) \\ &\leq 1. \end{aligned}$$

In particular, since $f \pm (1-\delta)g \neq f$, this contradicts the fact that f is extreme, so in particular, we must have $|f(x)| = 1$ for all $x \in \Omega$, yielding that Ω is compact. This gives (i).

Having seen that (i) holds, we have that $\operatorname{ext}\left(B_{C(\Omega)}\right)$ is contained in the set of all $u \in C(\Omega)$ with $|u| = 1$ on Ω. Our next task is to show that any such u is extreme.

Write $u = \frac{1}{2}(f+g)$ with $f, g \in B_{C(\Omega)}$. Then, $u(x) = \frac{1}{2}(f(x) + g(x))$ with $|f(x)|, |g(x)| \leq 1$ and $|u(x)| = 1$. Since the extreme points of the closed unit ball of $\mathbb{C}$ is S^1, it follows that $f(x) = g(x) = u(x)$ for all $x \in \Omega$. □

The most important result in extremal structure is the *Krein–Milman Theorem*.

Theorem: Let $C \subseteq X$ be a compact convex subset of a locally convex topological vector space. Then,

$$\overline{\operatorname{conv}}(\operatorname{ext}(C)) = C.$$

Proof. We start by showing that there are extreme points in C.

Let C_0 be a closed face of C (one may choose C for this closed face). Since a face is a face is a face, we may consider the set $\mathcal{C}$ of closed faces in C_0 ordered by containment. Notice that C is compact, so if $\{F_i\}_{i \in I}$ is a totally ordered subset of $\mathcal{C}$, then the finite intersection property is satisfied, so that $\bigcap_{i \in I} F_i$ is non-empty. By Zorn's Lemma, there is a minimal face which we will call F.

Let $\varphi \in X^*$ be any continuous linear functional. Set

$$\begin{aligned} s &= \inf\{\operatorname{Re}\langle x, \varphi\rangle : x \in F\} \\ F_\varphi &= \{x \in F : \operatorname{Re}\langle x, \varphi\rangle = s\}. \end{aligned}$$

First, we observe that F is compact and $x \mapsto \operatorname{Re}\langle x, \varphi\rangle$ is lower semicontinuous, so s is attained. In particular, this means $\emptyset \neq F_\varphi \subseteq F$. Therefore, by minimality, $F_\varphi = F$. Since φ is arbitrary, and X^* separates the points of X, it follows that $F = \{e_0\}$ for some element $e_0 \in C$. In particular, $\operatorname{ext}(C) \neq \emptyset$, and $\operatorname{ext}(C) \cap C_0 \neq \emptyset$.

Suppose there were $x \in C \backslash \overline{\operatorname{conv}}(\operatorname{ext}(C))$. Find an open convex neighborhood U with $x \in U$ and $U \cap \overline{\operatorname{conv}}(\operatorname{ext}(C)) = \emptyset$. By separating Hahn–Banach, there is $\varphi \in X^*$ and $t \in \mathbb{R}$ such that

$$\operatorname{Re}(\varphi(U)) < t \leq \operatorname{Re}\varphi(\overline{\operatorname{conv}}(\operatorname{ext}(C))).$$

In particular, if we let $s = \min\{\operatorname{Re}\langle x, \varphi\rangle : x \in C\}$, and define $C_\varphi = \{x \in C : \operatorname{Re}\langle x, \varphi\rangle = s\}$, then we have that $s < t$, and $C_\varphi \cap \operatorname{ext}(C) = \emptyset$, contradicting the definition of $\operatorname{ext}(C)$. □

One central application of the Krein–Milman Theorem is a proof of the Stone–Weierstrass Theorem.

Theorem: If X is compact, and $A \subseteq C(X)$ is a closed unital $*$-subalgebra that separates the points of X, then $A = C(X)$.

Proof. Suppose $A^\perp \neq 0$. Since $B_{A^\perp}$ is weak* compact, there is an extreme point μ of $B_{A^\perp}$. Let $K = \operatorname{supp}(\mu)$. Since $A^\perp \neq 0$, we may take $\|\mu\| = 1$ and we have that $K \neq \emptyset$. Let $x_0 \in K$. We will show that $K = \{x_0\}$.

Let $x \in X$ with $x \neq x_0$. Since A is self-adjoint, contains the constants, and separates points, we may assume that there is some $f_0 \in A$ with $0 = f_0(x) < f_0(x_0)$. Write $f = \frac{f_0}{\|f_0\|+1}$. Then, $f(x) = 0$ and $0 \leq f < 1$ on X. Since A is an algebra, gf and $g(1-f)$ are in A for each $g \in A$.

Since $\mu \in A^\perp$,

$$\begin{aligned} 0 &= \int gf \, d\mu \\ &= \int g(1-f)\, d\mu. \end{aligned}$$

In particular, this means that $f\mu$ and $(1-f)\mu$ are contained in $A^\perp$, where $f\mu$ and $(1-f)\mu$ denote measures whose Radon–Nikodym derivatives with respect to μ are given by f and $(1-f)$ respectively. Set

$$\begin{aligned} \alpha &= \|f\mu\| \\ &= \int |f| \, d|\mu|. \end{aligned}$$

There is an open neighborhood U of x_0 and some $\varepsilon > 0$ such that $f(y) > \varepsilon$ for all $y \in U$. Therefore,

$$\begin{aligned} \alpha &= \int f \, d|\mu| \\ &\geq \int_U f \, d|\mu| \\ &\geq \varepsilon |\mu|(U) \\ &> 0, \end{aligned}$$

where the latter follows from the fact that $U \cap K \neq \emptyset$.

Similarly, since $f(x_0) < 1$, it follows that $\alpha < 1$, meaning that $0 < \alpha < 1$. Similarly, we have that $1 - \alpha = \|(1-f)\mu\|$. Since

$$\mu = \alpha\left(\frac{f\mu}{\|f\mu\|}\right) + (1-\alpha)\left(\frac{(1-f)\mu}{\|(1-f)\mu\|}\right),$$

and μ is extreme, it follows that $\mu = \frac{f\mu}{\|f\mu\|}$, but this is equal to $\frac{1}{\alpha} f\mu$.

The only way that μ and $\frac{1}{\alpha} f\mu$ can be equal is if $\alpha^{-1} f = 1$ μ-a.e. Yet, this means that $f = \alpha$ on K. Since $\alpha = f(x_0) > f(x) = 0$, it follows that $x \notin K$. Thus, $K = \{x_0\}$, meaning that $\mu = \gamma \delta_{x_0}$ for some $\gamma \in S^1$.

Since $\mu \in A^\perp$, and $1 \in A$, it follows that $0 = \int 1 \, d\mu = \gamma$, which is a contradiction. Thus, $A^\perp = 0$, so $A = C(X)$. □

C^*-Algebras

The Structure of Abelian C^*-Algebras

Definition: If A is a (unital) abelian Banach algebra, then nonzero algebra homomorphisms $\alpha\colon A \to \mathbb{C}$ are known as characters.

Proposition: Let A be an abelian (unital) Banach algebra, and M a maximal ideal. Then, there is a homomorphism $h\colon A \to \mathbb{C}$ such that $M = \ker(h)$. Furthermore, if $h\colon A \to \mathbb{C}$ is a nonzero homomorphism, then $\ker(h)$ is a maximal ideal.

Proof. If M is a maximal ideal, then M is closed, so A/M is a unital Banach algebra. Let $\pi\colon A \to A/M$ be the corresponding quotient map. Suppose $a \in A$ and $\pi(a)$ is not invertible in A/M. Then, $\pi(Aa) = \pi(a)$ modulo A/M is a proper ideal. If $I = \pi^{-1}(\pi(Aa))$, then I is a proper ideal of A with $M \subseteq I$. Since M is maximal, $M = I$, so that $\pi(Aa) = \pi(M)$, meaning $\pi(a) = 0$.

Thus, A/M is a field. By the Gelfand–Mazur theorem, $A/M = \mathbb{C}$. If $\widetilde{h}\colon A/M \to \mathbb{C}$ is defined by $\widetilde{h}(\lambda + M) = \lambda$, then $h\colon A \to \mathbb{C}$ defined by $h = \widetilde{h} \circ \pi$ is a homomorphism with $\ker(h) = M$.

Conversely, if $h\colon A \to \mathbb{C}$ is a nonzero homomorphism, then $\ker(h) = M$ is a nontrivial closed ideal with $A/M \cong \mathbb{C}$, so M is maximal.

If h and h' are two such homomorphisms with $\ker(h) = \ker(h')$, then there is $\alpha \in \mathbb{C}$ such that $h = \alpha h'$, but this means $1 = h(1) = \alpha h'(1) = \alpha$, so that $h = h'$. □

Definition: Let A be an abelian (unital) Banach algebra. We let $\Sigma(A)$ be the set of all unital algebra homomorphisms equipped with the relative weak* topology as a subset of A^*. We call this the *character space* or *maximal ideal space*.

Theorem: If A is an abelian Banach algebra, then $\Sigma(A)$ is a compact Hausdorff space. Furthermore, if $a \in A$, then $\sigma(a) = \{h(a) : h \in \Sigma(A)\}$.

Proof. Since all unital Banach algebra homomorphisms have norm 1, it suffices to show that Σ is weak*-closed.

Let $(h_i)_i$ be a net in Σ, and let $h \in B_{A^*}$ be such that $h_i \xrightarrow{\text{wk}^*} h$. We observe that

$$\begin{aligned} h(ab) &= \lim_i h_i(ab) \\ &= \lim_i h_i(a)h_i(b) \\ &= h(a)h(b). \end{aligned}$$

Thus, h is a homomorphism, with $\lim_i h_i(1) = 1$, so that $h \in \Sigma$.

Suppose $h \in \Sigma$, and $\lambda = h(a)$. Then, $a - \lambda 1 \in \ker(h)$, so that $a - \lambda$ is not invertible and $\lambda \in \sigma(a)$. Now, if $\lambda \in \sigma(a)$, then $a - \lambda 1$ is not invertible, so $A(a - \lambda 1)$ is a proper ideal. Let M be a maximal ideal of A containing $A(a - \lambda 1)$. Then, $M = \ker(h)$ for some $h \in \Sigma(A)$, so that $\sigma(a) = \{h(a) : h \in \Sigma(A)\}$. □

Theorem: If X is compact, then $\Sigma(C(X)) \cong X$ via the identification $x \mapsto \delta_x$.

Proof. Let $\Delta\colon X \to \Sigma(C(X))$ be defined by $\Delta(x) = \delta_x$. Then, we have that $\Delta(X) \subseteq \Sigma(C(X))$, and Δ is a homeomorphism onto its image when the image is given the weak* topology.

Let $h \in \Sigma(C(X))$. Then, we have some measure μ such that

$$h(f) = \int f \, d\mu$$

with $\|\mu\| = 1$, and $\mu(X) = h(1) = 1$, so that $\mu \geq 0$. Let $x \in \operatorname{supp}(\mu)$.

If $M = \{f \in C(X) : f(x) = 0\}$, then M is a maximal ideal of $C(X)$. Our goal is to show that $\ker(h) \subseteq M$. Let $f \in \ker(h)$. Since $\ker(h)$ is an ideal, it follows that $|f|^2 \in \ker(h)$, so that

$$0 = \int |f|^2 \, d\mu,$$

whence $|f|^2 = 0$ μ-a.e., so since f is continuous, $f \equiv 0$ on $\operatorname{supp}(\mu)$. Thus, $f(x) = 0$, so $f \in M$. □

Definition: Let A be an abelian Banach algebra with character space $\Sigma(A)$. The *Gelfand Transform* of $a \in A$ is the map $\hat{a}\colon \Sigma(A) \to \mathbb{C}$ given by $\hat{a}(h) = h(a)$. We let $\Gamma(a) = \hat{a}$ denote the Gelfand transform on occasion.

Theorem: If A is an abelian Banach algebra with character space $\Sigma(A)$, then $\Gamma\colon A \to C(\Sigma(A))$ taking $a \mapsto \hat{a}$ is a norm 1 continuous algebra homomorphism of A with kernel given by

$$\bigcap \{M : M \text{ is a maximal ideal of } A\}.$$

Moreover, $\|\hat{a}\| = r(a)$, where $r(a)$ is the spectral radius.

Proof. If $(h_i)_i \to H$ in $\Sigma(A)$, then $h_i \xrightarrow{\text{wk}^*} h$ in A^*. Therefore, for any $a \in A$, we have

$$\begin{aligned} \hat{a}(h_i) &= h_i(a) \\ &\to h(a) \\ &= \hat{a}(h), \end{aligned}$$

so that $\hat{a} \in C(\Sigma(A))$.

Next, observe that

$$\begin{aligned} \Gamma(ab)(h) &= \widehat{ab}(h) \\ &= h(ab) \\ &= h(a)h(b) \\ &= \hat{a}(h)\hat{b}(h). \end{aligned}$$

Thus, $\Gamma(ab) = \Gamma(a)\Gamma(b)$. Moreover, since each character has norm 1, we have $|\hat{a}(h)| = |h(a)| \leq \|a\|$, so that $\|\Gamma(a)\| \leq \|a\|$, meaning Γ is continuous with norm at most 1. Since $\gamma(1) = 1$, it follows that $\left\|\gamma\right\| = 1$. □

Proposition: If A is a unital Banach algebra generated by $a \in A$, then there is a homeomorphism $\tau\colon \Sigma(A) \to \sigma(a)$ such that if $\gamma\colon A \to C(\Sigma(A))$ is the Gelfand transform, and p is a polynomial, then $\gamma(p(a)) = p \circ \tau = \tau^*(p)$.

Proof. We note that τ is surjective. If $\tau(h_1) = \tau(h_2)$, then $h_1(a) = h_2(a)$, so $h_1(a^n) = h_2(a^n)$ by multiplicativity and $h_1(p(a)) = h_2(p(a))$ by linearity. Since the set of polynomials in a is dense in A, and elements of $\Sigma(A)$ are dense, it follows that τ is injective. Since τ is an injective map from a compact space to a Hausdorff space, it follows that τ is a homeomorphism. □

The most important case study for the Gelfand transform is analyzing the group algebra of a locally compact group.

If G is a locally compact group, then we let $f, g \in L^1(G)$ be equipped with the convolution product

$$f * g(x) = \int f\left(xy^{-1}\right) g(y)\, dy,$$

where the integral is taken with respect to the Haar measure on G. If $L^1(G)$ has an identity under convolution, then this identity is necessarily given by the Dirac mass δ_e, but since δ_e is both in $L^1(G)$ and $L^\infty(G)$, it follows that $\delta_e * \delta_e = \delta_e$ is continuous, so that $\{e\}$ is a clopen set, implying that G is discrete. Analogously, if G is discrete, then δ_e is the identity. Conversely, if However, if G is not discrete, then $L^1(G)$ does not have an identity. However, $L^1(G)$ does have an approximate identity.

Theorem: Let G be a locally compact abelian group. If $\gamma\colon G \to S^1$ is a continuous homomorphism, we define

$$\hat{f}(\gamma) = \int f(x)\gamma\left(x^{-1}\right) dx$$

to be the Fourier transform of $f \in L^1(G)$. Then, for each γ, $f \mapsto \hat{f}(\gamma)$ is a nonzero homomorphism $L^1(G) \to \mathbb{C}$, and if $h\colon L^1(G) \to \mathbb{C}$ is a nonzero homomorphism, there is a continuous $\gamma\colon G \to S^1$ such that $h(f) = \hat{f}(\gamma)$.

Proof. If $f, g \in L^1(G)$, then relabeling as necessary, we have

$$\begin{aligned}\widehat{f * g}(\gamma) &= \int (f * g)(x)\gamma\left(x^{-1}\right) dx \\ &= \int \left(\int f\left(xy^{-1}\right)g(y)\, dy \right)\gamma\left(x^{-1}\right) dx \\ &= \int \left(\int f\left(xy^{-1}\right)\gamma\left(\left(xy^{-1}\right)^{-1}\right) dx \right) g(y)\gamma\left(y^{-1}\right) dy.\end{aligned}$$

By the invariance of the Haar integral, we have that

$$\widehat{f * g}(\gamma) = \hat{f}(\gamma)\hat{g}(\gamma).$$

Thus, $f \mapsto \hat{f}(\gamma)$ is a homomorphism. Since $\gamma(G) \subseteq S^1$, it follows that $\gamma \in L^\infty(G)$ and $\left\|\gamma\right\| = 1$, meaning that $f \mapsto \hat{f}(\gamma)$ is not identically zero.

Suppose $h\colon L^1(G) \to \mathbb{C}$ is a nonzero homomorphism. Since h is a bounded linear functional, then there is some ψ such that

$$h(f) = \int f(x)\psi(x)\, dx$$

with $\left\|\phi\right\|_{L^\infty} = \|h\| = 1$. We have, by using Fubini's theorem,

$$\begin{aligned}h(f * g) &= \int (f * g)(x)\psi(x)\, dx \\ &= \int \int g(y) f\left(y^{-1}x\right)\psi(x)\, dy\, dx \\ &= \int g(y) \int f\left(y^{-1}x\right)\psi(x)\, dx\, dy \\ &= \int g(y) h\left(L_y f\right) dy,\end{aligned}$$

where $L_y f(x) = f\left(y^{-1}x\right)$. Yet, we also have

$$\begin{aligned}h(f * g) &= h(f)h(g) \\ &= \int g(y)h(f)\psi(y)\, dy,\end{aligned}$$

so that

$$0 = \int g(y)\left(h\left(L_y f\right) - h(f)\psi(y)\right) dy$$

for all $g \in L^1(G)$. Since $y \mapsto h\left(L_y f\right) - h(f)\psi(y)$ is an element of $L^\infty(G)$, so that $h\left(L_y f\right) = h(f)\psi(y)$ for almost every $y \in G$. Choose $f \in L^1(G)$ such that $h(f) \neq 0$. Then, $\psi(y) = \frac{h\left(L_y f\right)}{h(f)}$ almost everywhere. The right-hand side of this equation is continuous in y, so we may assume that $h\left(L_y f\right) = h(f)\psi(y)$ holds everywhere. □

We note that this means we may identify $\Sigma\left(L^1(G)\right)$ with the Pontryagin dual $\hat{G}$. It is equipped with the

weak*-topology (i.e., pointwise convergence).

Theorem: If $y \in \mathbb{R}$, then $\gamma_y(x) = e^{ixy}$ defines a character on $\mathbb{R}$, and all characters on $\mathbb{R}$ are of this form. For any $y \in \mathbb{R}$ and $f \in L^1(\mathbb{R})$, one has that

$$\hat{f}(y) = \int f(x)e^{-ixy}\,dx.$$

Proof. If $y \in \mathbb{R}$, then $\left|\gamma_y(x)\right| = 1$ for all x, and $\gamma_y(x_1 + x_2) = \gamma_y(x_1)\gamma_y(x_2)$. In particular, $\gamma_y \in \hat{\mathbb{R}}$, and $\gamma_{y_1+y_2}(x) = \gamma_{y_1}(x)\gamma_{y_2}(x)$, and so $y \mapsto \gamma_y$ is a homomorphism of $\mathbb{R}$ into $\hat{\mathbb{R}}$.

Now, suppose $\gamma \in \hat{\mathbb{R}}$. Then, $\gamma(0) = 1$, so there is some $\delta > 0$ such that

$$\int_0^\delta \gamma(x)\,dx = a,$$

with $a \neq 0$. In particular, we have

$$\begin{aligned} a\gamma(x) &= \gamma(x)\int_0^\delta \gamma(t)\,dt \\ &= \int_0^\delta \gamma(x+t)\,dt \\ &= \int_x^{x+\delta} \gamma(t)\,dt. \end{aligned}$$

So, we have that $\gamma(x) = a^{-1}\int_x^{x+\delta}\gamma(t)\,dt$. Since γ is continuous, one has that γ is differentiable, and

$$\frac{\gamma(x+h)-\gamma(x)}{h} = \gamma(x)\left(\frac{\gamma(h)-1}{h}\right),$$

meaning that $\gamma'(x) = \gamma(0)\gamma(x)$. Since $\gamma(0) = 1$ and $\left|\gamma(x)\right| = 1$ for all x, we have that $\gamma = \gamma_y$ for some $y \in \mathbb{R}$. □

Continuous Functional Calculus

Proposition: If A is a C^*-algebra, and $h\colon A \to \mathbb{C}$ is a nonzero homomorphism, then $h(a) \in \mathbb{R}$ whenever $a \in A_{\text{s.a.}}$.

Proof. Assume A has an identity. For any $t \in \mathbb{R}$, we have

$$\begin{aligned} |h(a+it1)| &\leq \|a+it1\|^2 \\ &= \left\|a^2+t^2 1\right\| \\ &\leq \|a\|^2 + t^2. \end{aligned}$$

If $h = x + iy$ with $x, y \in \mathbb{R}$, then

$$\begin{aligned} \|a\|^2 + t^2 &\geq \left|\alpha + i(\beta+t)\right|^2 \\ &= \alpha^2 + (\beta+t)^2 \\ &= \alpha^2 + \beta^2 + 2\beta t + t^2. \end{aligned}$$

If it were the case that $\beta \neq 0$, then $t \to \pm\infty$ would yield a contradiction given the sign of β. So, $\beta = 0$, hence $h(a) \in \mathbb{R}$. □

Theorem: If A is a unital abelian C^*-algebra, and $\Sigma(A)$ is its character space, then the Gelfand transform $\Gamma\colon A \to C(\Sigma(A))$ is an isometric $*$-isomorphism.

Proof. If $a \in A$ and $h \in \Sigma(A)$, then $\widehat{a^*}(h) = h(a^*) = \overline{\hat{a}(h)}$, whence $\widehat{a^*} = \overline{\hat{a}}$.

We have that $\|\hat{x}\| \leq \|x\|$ for every $x \in A$. Since $\|\hat{x}\|$ is the spectral radius of x, it follows that $\|x^*x\| = \|\hat{x}^*\hat{x}\|$ for all $x \in A$, but this means that $\|x\|^2 = \|\hat{x}\|^2$ for all $x \in A$, whence Γ is an isometry.

Since Γ is an isometry, it has closed range, so it suffices to show it is dense. Since $\Gamma(1) = 1$, it follows that $\Gamma(A)$ contains the constants. Since Γ preserves complex conjugation and separates the points of $\Sigma(A)$, we get the desired result. □

If $a \in A$ is a normal element, we let $E = C^*(a)$. Then, the map $\mathrm{ev}_a \colon \Sigma(E) \to \sigma(a)$ be the evaluation map $h \mapsto h(a)$. Since this is a homeomorphism, as it is a continuous bijection from a compact space to a Hausdorff space where we recognize that the value of h is uniquely determined by its value on a. Gelfand duality gives an isometric $*$-isomorphism $\mathrm{ev}_a^* \colon C(\sigma(a)) \to C(\Sigma(E))$, given by $\mathrm{ev}_a^*(f) = f \circ \mathrm{ev}_a$.

The Gelfand transform gives an isometric $*$-isomorphism $\Gamma \colon E \to C(\Sigma(E))$, so the composition $\Gamma_a := \Gamma^{-1} \circ \mathrm{ev}_a^*$ from $C(\sigma(a)) \to E$. In particular, this gives us the *continuous functional calculus* for any function $f \in C(\sigma(a))$, and that $C^*(a) \cong C(\sigma(a))$.

In particular, we get the *spectral mapping property*, which says that for any $a \in A$, we have that $f(\sigma(a)) = \sigma(f(a))$, since $\Gamma_a \colon C(\sigma(a)) \to C^*(a)$ is an isomorphism, so that spectra are preserved, and we recall that $\sigma(f) = \mathrm{range}(f)$ for all $f \in C(\sigma(a))$.

Ordering and Positivity

Definition: If A is a C^*-algebra, we say $a \in A$ is *positive* if $a \in A_{\text{s.a.}}$ and $\sigma(a) \subseteq [0, \infty)$. If a is positive, then we write $a \geq 0$. We let A_+ be the set of all positive elements of A.

Theorem: For any $a \in A_{\text{s.a.}}$, there are positive elements $u, v \in A_+$ such that $a = u - v$ and $uv = 0$. Such a decomposition is unique. We call this the orthogonal decomposition.

Proof. Using functional calculus on $\sigma(a) \subseteq \mathbb{R}$, take

$$f_1(t) = \max(0, t)$$
$$f_2(t) = -\min(0, t),$$

so that $u = f_1(a)$ and $v = f_2(a)$.

The proof of uniqueness follows from examining the uniqueness of the C^*-algebra $C^*(a, u, v, u_1, v_1)$ if $u_1 - v_1 = a$ is another orthogonal decomposition. □

Theorem: The following are equivalent:

(i) $a \geq 0$;

(ii) $a = b^2$ for some $b \in A_{\text{s.a.}}$;

(iii) $a = x^*x$ for some $x \in A$;

(iv) $a \in A_{\text{s.a.}}$ and $\|t1 - a\| \leq t$ for all $t \geq \|a\|$;

(v) $a \in A_{\text{s.a.}}$ and $\|t1 - a\|$ for some $t \geq \|a\|$.

Proof. We start by showing that (v) implies (i). Since $a \in A_{\text{s.a.}}$, we have that $C^*(a)$ is abelian, and $X = \sigma(a)$ is contained in $\mathbb{R}$. Then, $f \mapsto f(a)$ is a $*$-isomorphism of $C(X)$ onto $C^*(a)$. Then, assuming the isomorphism, we have $|t - x| \leq t$ for some $t \geq \|a\| = \sup\{|s| : s \in \sigma(a)\}$. We observe then that $x \geq 0$ for all $x \in X$, so that $\sigma(a) \subseteq [0, \infty)$.

Analogously, if $a \geq 0$, then the isomorphism $C^*(a) \cong C(\sigma(a))$ yields that $|t - x| \leq t$ for all $t \geq \sup\{x : x \in \sigma(a)\}$, so that we get that $\|t1 - a\| \leq t$ for all $t \geq \|a\|$, giving (i) implies (iv).

Next, assuming (iii), we have that if $a = x^*x$, then $a \in A_{\text{s.a.}}$. Write $a = u - v$ with $u, v \geq 0$ and $uv = vu = 0$. Write

$$xv^{1/2} = b + ic$$

for some $b, c \in A_{\text{s.a.}}$. Then, $\left(xv^{1/2}\right)^*\left(xv^{1/2}\right) = b^2 + c^2 - i(bc - cb)$. At the same time, we have $\left(xv^{1/2}\right)^*\left(xv^{1/2}\right) = v^{1/2}x^*xv^{1/2} = -v^2$. Thus, $i(bc - cb) = -v^2 - b^2 - c^2$. In particular, since A_+ is a closed cone (see below), we have that $i(bc - cb) \leq 0$. Analogously, we have that $\left(xv^{1/2}\right)^*\left(xv^{1/2}\right) = -v^2 \leq 0$, and that $-y \coloneqq \left(xv^{1/2}\right)\left(xv^{1/2}\right)^* \leq 0$, where $y \in A_+$. Therefore, we have $-y = (b + ic)(b - ic) = b^2 + c^2 - i(bc - cb)$. Thus, we have

$$\begin{aligned} i(bc - cb) &= b^2 + c^2 + y \\ &\in A_+, \end{aligned}$$

meaning that $i(bc - cb) = (-A_+) \cap A_+ = 0$. Yet, this means that $-v^2 \in A_+ \cap (-A_+)$, so $v^2 = 0$. Since $v \leq 0$, we have that $v = 0$, so that $a = u \geq 0$. □

Proposition: If A is a C^*-algebra, then A_+ is a closed cone.

Proof. Let $(a_n)_n \subseteq A_+$. Suppose $a_n \to a$. We have that $a \in A_{\text{s.a.}}$. Since $\|a_n - \|a_n\|1\| \leq \|a_n\|$, it follows that $\|a - \|a\|1\| \leq \|a\|$, so that $a \geq 0$.

We have that $\lambda a \geq 0$ whenever $a \geq 0$ and $\lambda \geq 0$. If $a, b \in A_+$, we will show that $a + b \geq 0$. It suffices to assume that $\|a\|, \|b\| \leq 1$. Since

$$\begin{aligned} \left\| 1 - \frac{1}{2}(a + b) \right\| &= \frac{1}{2}\|(1 - a) + (1 - b)\| \\ &\leq 1, \end{aligned}$$

it follows that $\frac{1}{2}(a + b) \geq 0$.

If $a \in A_+ \cap (-A_+)$, then $a = a^*$ and $\sigma(a) = \{0\}$. Since $\|a\| = r(a)$, it follows that $a = 0$. □

We recall that the definition of the absolute value of an element is given by $|a| = (a^*a)^{1/2}$. The most immediate application of the continuous functional calculus applied to operators on a Hilbert space is the polar decomposition.

Theorem (Polar Decomposition)**:** If $A \in B(H)$, then there is a partial isometry W with initial space $\ker(A)^\perp$ and final space $\overline{\text{range}}(A)$ such that $A = W|A|$. This decomposition is also unique.

Proof. We have that $\|Ah\|^2 = \||A|h\|^2$. Since $\text{range}(A^*)^\perp = \ker(A)$, it follows that $\text{range}(A^*) \subseteq \ker(A)^\perp$ is dense. If $f \in \text{range}(A^*)$, then $f = A^*g$ for some $g \in \overline{\text{range}}(A)$, so that $\{A^*Ak : k \in H\}$ is dense in $\overline{\text{range}}(A^*) = (\ker(A))^\perp$.

Since $A^*Ak = |A|^2k$, it follows that $\{|A|h : h \in H\}$ is dense in $\ker(A)^\perp$. If $W = \text{range}(|A|) \to \text{range}(A)$ is defined by $W(|A|h) = Ah$, then W is a well-defined isometry that extends to $\ker(A)^\perp$. Defining W to be 0 on $\ker(A)$ yields a partial isometry with initial space $\ker(A)^\perp$ and final space $\overline{\text{range}}(A)$. □

Proposition: If $\rho\colon A \to B$ is an injective $*$-homomorphism of (unital) C^*-algebras, then ρ is an isometry.

Proof. Let ρ be injective. Let $a \in A$ with $a = a^*$. Then, $\rho(a) = \rho(a)^*$ and $\sigma(\rho(a)) \subseteq \sigma(a)$. Suppose $\sigma(\rho(a)) \neq \sigma(a)$. Then, by Urysohn's Lemma, we may find some function f on $\sigma(\rho(a))$ such that $f(t) = 0$ on $\sigma(\rho(a))$ but $f \neq 0$ on $\sigma(a)$. In particular, by continuous functional calculus, $f(\rho(a)) = 0$ and $f(a) \neq 0$.

Let $(p_n)_n$ be polynomials uniformly converging to f on $\sigma(a)$. Thus, $p_n(a) \to f(a)$ and $p_n(\rho(a)) \to f(\rho(a))$. Since $p_n(\rho(a)) = \rho(p_n(a)) \to \rho(f(a))$, it follows that $\rho(f(a)) = f(\rho(a)) = 0$. However, ρ was assumed injective, so it follows that $\sigma(a) = \sigma(\rho(a))$ if $a = a^*$. Thus, a and $\rho(a)$ have equal spectral radii, so by using the C^*-identity, we obtain the desired result. □

The GNS Representation

Proposition: Suppose f is a positive linear functional on a C^*-algebra A. Then,

$$\left|f(y^*x)\right|^2 \leq f(y^*y)f(x^*x).$$

Proof. For any $x, y \in A$, we may define a positive sesquilinear form $[x, y] = f(y^*x)$. The Cauchy–Schwarz inequality thus applies. □

Theorem (GNS Construction)**:** Let A be a unital C^*-algebra. If φ is a positive linear functional on A, then there is a cyclic representation $\big(\lambda_\varphi, L^2(A,\varphi)\big)$ for A with cyclic vector e such that $\varphi(a) = \big\langle \lambda_\varphi(a)e, e\big\rangle$.

Proof. Let φ be a positive linear functional on A, and let $N = \{x \in A : \varphi(x^*x) = 0\}$. Then, N is a closed (since φ is continuous) left ideal in A, where the fact that N is a left ideal emerges from the fact that for all $x \in N$ and $a \in A$,

$$\begin{aligned}\big|\varphi((ax)^*(ax))\big|^2 &= \big|\varphi(x^*(a^*ax))\big|^2 \\ &\leq \varphi(x^*x)\varphi(x^*a^*aa^*ax) \\ &= 0.\end{aligned}$$

We will write $\hat{x} := x + N$ whenever $x \in A$. Thus, we may view the vector space A/N to be equipped with the inner product $\langle \hat{x}, \hat{y}\rangle = \varphi(y^*x)$. We will let $L^2(A,\varphi)$ denote the completion of A/N with respect to the norm defined by this inner product.

Since N is a left ideal, we have that $\hat{x} \mapsto \widehat{ax}$ is a well-defined linear transformation satisfying

$$\begin{aligned}\|\widehat{ax}\|^2 &= \langle \widehat{ax}, \widehat{ax}\rangle \\ &= f(x^*a^*ax).\end{aligned}$$

Since $\|a^*a\|1 - a^*a \geq 0$, it follows that $0 \leq x^*(\|a^*a\|1 - a^*a)x$, so that $x^*a^*ax \leq \|a\|^2x^*x$. In particular, we have that $\|\widehat{ax}\|^2 \leq \|a\|^2 f(x^*x) = \|a\|^2\|\hat{x}\|^2$. In particular, we may extend the map $\lambda(a)(\hat{x}) = \widehat{ax}$ to $\lambda_\varphi \in B\big(L^2(A,\varphi)\big)$.

If we set $e = \hat{1}$, then $\hat{A} := \{\hat{a} : a \in A\}$ is dense in $L^2(A,\varphi)$, so e is a cyclic vector for λ_φ. □

Remark: If $\tau\colon A \to \mathbb{C}$ is a trace (rather than simply a positive linear functional or a state), then the left-ideal N is a two-sided ideal, so we are automatically equipped with a right action $\rho_\tau\colon A^{\mathrm{op}} \to B\big(L^2(A,\tau)\big)$ such that $\rho_\tau(a^{\mathrm{op}})\hat{x} = \widehat{xa}$.

Proposition: If A is a unital C^*-algebra, then for any $a \in A_+$, we have

$$\|a\| = \sup\{\varphi(a) : \varphi \in S(A)\}.$$

Proof. When $A = C(X)$ with X compact, and $f \in C(X)_+$, we have some $x \in X$ such that

$$\begin{aligned}\|f\| &= f(x) \\ &= \int f\, d\delta_x \\ &= \sup\left\{\int f\, d\mu : \mu \in \mathrm{Prob}(X)\right\}.\end{aligned}$$

Applying this argument to $C^*(a)$ for a positive yields a state φ_0. By the Hahn–Banach theorem, there is a norm-one linear functional $\varphi\colon A \to \mathbb{C}$ such that $\varphi|_{C^*(a)} = \varphi_0$. Since $1 \in C^*(a)$, it follows that $\varphi(1) = 1 = \big\|\varphi\big\|$, so that φ is a state.

We may use the same process for self-adjoints and the C^*-identity to obtain the desired result. □

Corollary: If A is a C^*-algebra, then A admits a faithful representation.

Proof. Consider the direct sum

$$H = \bigoplus_{\varphi \in S(A)} L^2(A,\varphi).$$

The corresponding representation $a \mapsto \big(\hat{a}_\varphi\big)_{\varphi \in S(A)}$ is then injective since there is some $\varphi \in S(A)$ such that $\hat{a}_\varphi \neq 0$. □

Note that the state space $S(A)$ is a weak*-compact convex subset of A^*, so it has extreme points that are

known as pure states.

Theorem: Let ρ be a state. Then, the GNS representation π_ρ is irreducible if and only if ρ is a pure state.

Proof. Let $\pi = \pi_\rho$ be not irreducible. Then, there is an invariant subspace K of $L^2(A, \rho)$ such that $P = P_K$ satisfies $P \neq 0 \neq 1-P$. We will write ρ as a nontrivial convex combination of states. Since K is invariant, we have for any $a \in A$ that $\pi(a)P = P\pi(a)$. In particular, we also have that $\left\|P\xi_\rho\right\| \neq 0$, and similarly, $\left\|(1-P)\xi_\rho\right\| \neq 0$. Therefore, $\lambda = \left\|P\xi_\rho\right\|^2$ satisfies $0 < \lambda < 1$, and

$$\begin{aligned}\phi(a) &= \frac{1}{\lambda}\left\langle \pi(a)P\xi_\rho, P\xi_\rho \right\rangle \\ \psi(a) &= \frac{1}{1-\lambda}\left\langle \pi(a)(1-P)\xi_\rho, (1-P)\xi_\rho \right\rangle.\end{aligned}$$

Therefore, we have

$$\begin{aligned}\rho(a) &= \left\langle \pi(a)\xi_\rho, \xi_\rho \right\rangle \\ &= \left\langle \pi(a)P\xi_\rho, P\xi_\rho \right\rangle + \left\langle \pi(a)(1-P)\xi_\rho, (1-P)\xi_\rho \right\rangle \\ &= \lambda\phi(a) + (1-\lambda)\psi(a).\end{aligned}$$

We note that if it were the case that $\phi = \rho$, then we would have $\left\|P\xi_\rho\right\|^2 \xi_\rho = P\xi_\rho$, which would imply that $\left\|P\xi_\rho\right\|^2 = 1$, contradicting the fact that $0 < \lambda < 1$.

Now, suppose ϕ □

Spectral Theory

Spectral theory is primarily concerned with representations of abelian C^*-algebras. Consider a normal operator N on H. Then, $C^*(N)$ is an abelian C^*-algebra, and the continuous functional calculus $f \mapsto f(N)$ yields a $*$-isomorphism of $C(\sigma(N))$ onto $C^*(N)$. In particular, $f \mapsto f(N)$ is a representation of the abelian C^*-algebra $C^*(N)$.

We will find a way to extend this further, as well as find some useful applications to the structure of abelian von Neumann algebras and understanding unitary invariants for normal operators.

Definition: Let X be a set, Σ a σ-algebra on X, and H a Hilbert space. A *spectral measure* for (X, Σ, H) is a map $E\colon \Sigma \to B(H)$ satisfying

(a) for each $\Delta \in \Sigma$, $E(\Delta)$ is a projection;

(b) $E(\emptyset) = 0$ and $E(X) = 1$;

(c) $E(\Delta_1 \cap \Delta_2) = E(\Delta_1)E(\Delta_2)$ for any $\Delta_1, \Delta_2 \in \Sigma$;

(d) given pairwise disjoint $\{\Delta_n\}_{n=1}^\infty$, we have

$$E\left(\bigcup_{n=1}^\infty \Delta_n\right) = \sum_{n=1}^\infty E(\Delta_n),$$

where the latter is an SOT-convergent sum of pairwise orthogonal projections converging to the projection onto the supremum $\bigwedge_{n=1}^\infty E(\Delta_n)H$.

Recall that the SOT and WOT are defined by the seminorms

- $p_h(A) = \|Ah\|$ for the SOT, and
- $p_{h,k}(H) = |\langle Ah, k\rangle|$ for the WOT.

Example: If (X,Ω,μ) is a σ-finite measure space, and $\phi \in L^\infty(X,\mu)$, then $M_\phi \in B\Big(L^2(X,\mu)\Big)$ can be defined by $M_\phi(f) = \phi f$.

A net $(\phi_i)_{i\in I}$ in $L^\infty(X,\mu)$ is w^*-convergent to ϕ if and only if $M_{\phi_i} \to M_\phi$ in WOT. This follows from the fact that if $f,g \in L^2(X,\mu)$, then $f\overline{g} \in L^1(X,\mu)$, where we have $L^1(X,\mu)^* = L^\infty(X,\mu)$, so that if $(\phi_i)_i \to \phi$ weak*, then

$$\begin{aligned}\Big\langle M_{\phi_i} f, g\Big\rangle &= \int \phi_i (f\overline{g})\, d\mu \\ &\to \int \phi f \overline{g}\, d\mu \\ &= \Big\langle M_\phi f, g\Big\rangle,\end{aligned}$$

while the converse emerges from the fact that any element $f \in L^1(X,\mu)$ can be written as $f = g_1\overline{g_2}$.

Example: If E is a spectral measure for (X,Ω,H), then $E^{(n)}(\Delta) = E(\Delta)^{(n)}$ (under the diagonal embedding) is a spectral measure for $\Big(X,\Omega,H^{(n)}\Big)$.

We will now use spectral measures to define a representation of a C^*-algebra.

Lemma: Let E be a spectral measure on (X,Ω,H), and let $g,h \in H$. Then,

$$E_{g,h}(\Delta) = \langle E(\Delta)g, h\rangle$$

defines a countably additive measure on Ω with total variation bounded above by $\|g\|\|h\|$.

Proof. For countable additivity, we note the classic result that a sequence of projections $(P_n)_n \to P$ converges in WOT if and only if it converges in SOT. In particular, this means that if $\{\Delta_n\}_{n\in\mathbb{N}}$ is a sequence of disjoint subsets of X, then we have a sequence of projections

$$\left(\sum_{k=0}^{n} E(\Delta_k)\right)_n \to \sum_{k=0}^{\infty} E(\Delta_k)$$

converging in SOT, hence converging in WOT. Yet, this means that

$$\begin{aligned}\left\langle\left(\sum_{k=0}^{n} E(\Delta_k)\right)g, h\right\rangle &= \sum_{k=0}^{n}\langle E(\Delta_k)g,h\rangle \\ &\to \left\langle\left(\sum_{k=0}^{\infty} E(\Delta_k)\right)g, h\right\rangle,\end{aligned}$$

so that

$$\sum_{k=0}^{\infty}\langle E(\Delta_k)g,h\rangle = \left\langle E\left(\bigcup_{k=0}^{\infty}\Delta_k\right)g, h\right\rangle,$$

yielding our desired result.

If we have a finite set $\Delta_1,\dots,\Delta_n$ of pairwise disjoint sets, we choose $|\alpha_j| = 1$ such that $\Big|\Big\langle E\big(\Delta_j\big)g,h\Big\rangle\Big| = \alpha_j\Big\langle E\big(\Delta_j\big)g,h\Big\rangle$, whence

$$\begin{aligned}\sum_{j=1}^{n}\Big|\mu\big(\Delta_j\big)\Big| &= \left\langle\sum_{j=1}^{n}\alpha_j E\big(\Delta_j\big)g, h\right\rangle \\ &\le \left\|\sum_{j=1}^{n} E\big(\Delta_j\big)\alpha_j g\right\|\|h\|.\end{aligned}$$

Notice that the family of $E\big(\Delta_j\big)$ are pairwise orthogonal projections, meaning that by Pythagoras's Theorem and the additivity of the spectral measure, we have

$$\begin{aligned}\left\|\sum_j^n E\big(\Delta_j\big)\alpha_j g\right\|^2 &= \sum_{j=1}^n \left\|E\big(\Delta_j\big)g\right\|^2 \\ &= \left\|E\left(\bigcup_{j=1}^n \Delta_j\right)g\right\|^2 \\ &\leq \|g\|^2,\end{aligned}$$

whence $\sum_{j=1}^n \left|\mu\big(\Delta_j\big)\right| \leq \|g\|\|h\|$. Since this holds for any finite collection of subsets, we have $\left\|\mu\right\|_{\text{TV}} \leq \|g\|\|h\|$. □

We next concern ourselves with establishing "integration" with respect to the spectral measure.

Proposition: Let E be a spectral measure for (X, Ω, H), and let $\phi\colon X \to \mathbb{C}$ be a bounded Ω-measurable function. Then, there is a unique operator $A \in B(H)$ such that for any $\varepsilon > 0$ and partition $\{\Delta_1, \ldots, \Delta_n\}$ with $\sup_{x,x' \in \Delta_k} \left|\phi(x) - \phi(x')\right| < \varepsilon$ for each k, we have

$$\left\|A - \sum_{k=1}^n \phi(x_k) E(\Delta_k)\right\| < \varepsilon$$

for any $x_k \in \Delta_k$.

Proof. For each $g, h \in H$, define a (bounded) sesquilinear form

$$B(g,h) = \int \phi \, dE_{g,h},$$

with $|B(g,h)| \leq \left\|\phi\right\|_{L^\infty} \|g\|\|h\|$. Then, by the structure of bounded sesquilinear forms, there is a unique operator A such that $B(g,h) = \langle Ag, h\rangle$ for all $g, h \in H$.

Consider a partition $\{\Delta_1, \ldots, \Delta_n\}$ satisfying the conditions in the proposition. For any arbitrary vectors $g, h \in H$ and $x_k \in \Delta_k$, we have

$$\begin{aligned}\left|\langle Ag, h\rangle - \sum_{k=1}^n \phi(x_k)\langle E(\Delta_k)g, h\rangle\right| &= \left|\sum_{k=1}^n \int_{\Delta_k} (\phi(x) - \phi(x_k)) \, dE_{g,h}\right| \\ &\leq \sum_{k=1}^n \int_{\Delta_k} \left|\phi(x) - \phi(x_k)\right| d\left|E_{g,h}\right| \\ &\leq \varepsilon \int d\left|E_{g,h}\right| \\ &\leq \varepsilon \|g\|\|h\|.\end{aligned}$$

This yields the desired result. □

For the operator A in the above statement, we write

$$A = \int \phi \, dE.$$

That is, for any $g, h \in H$ and bounded Ω-measurable function ϕ on X, we have

$$\left\langle \left(\int \phi \, dE \right) g, h \right\rangle = \int \phi \, dE_{g,h}.$$

If $B_\infty(X,\Omega)$ denotes the set of bounded Ω-measurable functions $\phi\colon X\to\mathbb{C}$ with the norm $\|\phi\| = \sup_{x\in X}|\phi(x)|$, then $B_\infty(X,\Omega)$ is a unital Banach algebra — in fact, it is an abelian C^*-algebra.

Proposition: Suppose E is a spectral measure for (X,Ω,H), and $\rho\colon B_\infty(X,\Omega)$ is defined by taking $\rho(\phi) = \int \phi\, dE$.

Then, ρ is a representation of $B_\infty(X,\Omega)$ and $\rho(\phi)$ is a normal operator for every $\phi\in B_\infty(X,\Omega)$.

Proof. We will only focus on showing that ρ is multiplicative. Suppose $\phi,\psi\in B_\infty(X,\Omega)$. Let $\varepsilon>0$, and let $\{\Delta_1,\dots,\Delta_n\}$ be a Borel partition of X satisfying

$$\sup\Big\{\big|\omega(x)-\omega(x')\big| : x,x'\in\Delta_k, \omega=\phi,\psi,\phi\psi\Big\}<\varepsilon.$$

In particular, we have that for any $x_k\in\Delta_k$ and $\omega=\phi,\psi,\phi\psi$, we have

$$\left\|\int \omega\, dE - \sum_{k=1}^n \omega(x_k)E(\Delta_k)\right\| < \varepsilon.$$

The triangle inequality then gives

$$\begin{aligned}\left\|\int \phi\psi\, dE - \left(\int \phi\, dE\right)\left(\int \psi\, dE\right)\right\| &\le \varepsilon + \left\|\sum_{k=1}^n \phi(x_k)\psi(x_k)E(\Delta_k) - \left(\sum_{i=1}^n \phi(x_i)E(\Delta_i)\right)\left(\sum_{j=1}^n \psi\big(x_j\big)E\big(\Delta_{x_j}\big)\right)\right\| \\ &\quad + \left\|\left(\sum_{i=1}^n \phi(x_i)E(\Delta_i)\right)\left(\sum_{j=1}^n \psi\big(x_j\big)E\big(\Delta_j\big)\right) - \left(\int \phi\, dE\right)\left(\int \psi\, dE\right)\right\|.\end{aligned}$$

Since $E(\Delta_i)E\big(\Delta_j\big) = E\big(\Delta_i\cap\Delta_j\big)$, and $\{\Delta_1,\dots,\Delta_n\}$ is a partition, we have that the second term on the right-hand side is zero. Thus, we have

$$\begin{aligned}\left\|\int \phi\psi\, dE - \left(\int \phi\, dE\right)\left(\int \psi\, dE\right)\right\| &\le \varepsilon + \left\|\left(\sum_{i=1}^n \phi(x_i)E(\Delta_i)\right)\left(\sum_{j=1}^n \psi\big(x_j\big)E\big(\Delta_j\big) - \int \psi\, dE\right)\right\| \\ &\quad + \left\|\left(\sum_{i=1}^n \phi(x_i)E(\Delta_i) - \int \phi\, dE\right)\left(\int \psi\, dE\right)\right\| \\ &\le \varepsilon\big(1 + \|\phi\| + \|\psi\|\big).\end{aligned}$$

Thus, we get multiplicativity. □

Corollary: If X is a compact Hausdorff space, and E is a spectral measure on H defined on the Borel sets of X, then $\rho\colon C(X)\to B(H)$ given by $\rho(u) = \int u\, dE$ is a representation of $C(X)$.

Thus far, all we have obtained is a variety of the properties of a spectral measure. However, we do not know if such a measure exists. The following is the proof that it does.

Theorem: Suppose $\rho\colon C(X)\to B(H)$ is a representation. Then, there is a unique spectral measure E defined on the Borel subsets of X such that for all $g,h\in H$, $E_{g,h}$ is a regular measure, and we have

$$\rho(u) = \int u\, dE$$

for all $u\in C(X)$.

Theorem: Let $g,h\in H$. The map $u\mapsto\langle\rho(u)g,h\rangle$ is a linear functional on $C(X)$ with norm at most $\|g\|\|h\|$.

So, there is a unique measure $\mu_{g,h}$ in $M(X)$ such that

$$\langle \rho(u)g, h\rangle = \int u \, d\mu_{g,h}.$$

The pairing $(g,h) \mapsto \mu_{g,h}$ is sesquilinear, and for fixed $\phi \in B_\infty(X)$, the form

$$[g,h] = \int \phi \, d\mu_{g,h}$$

is a sesquilinear form with $|||[g,h]||| \leq \left\|\phi\right\| ||g|| ||h||$. Therefore, there is a unique bounded operator A such that $[g,h] = \langle Ag, h\rangle$ with $\|A\| \leq \left\|\phi\right\|$. We will denote this operator A by $\widetilde{\rho}(\phi)$. We obtain a well-defined map $\widetilde{\rho}\colon B_\infty(X) \to B(H)$ satisfying $\left\|\widetilde{\rho}(\phi)\right\| \leq \left\|\phi\right\|$ such that

$$\langle \widetilde{\rho}(\phi)g, h\rangle = \int \phi \, d\mu_{g,h}.$$

We start by claiming that $\widetilde{\rho}\colon B_\infty(X) \to B(H)$ is a representation with $\widetilde{\rho}|_{C(X)} = \rho$. That $\widetilde{\rho}$ restricted to $C(X)$ is equal to ρ follows directly from the definition of $\widetilde{\rho}$.

Now, let $\phi \in B_\infty(X)$. Consider ϕ as an element of $M(X)^* = C(X)^{**}$ by viewing it as corresponding to the linear functional

$$\mu \mapsto \int \phi \, d\mu.$$

By the Banach–Alaoglu theorem, we have that $\left\{u \in C(X) : \|u\| \leq \left\|\phi\right\|\right\}$ is weak*-dense in $\left\{L \in M(X)^* : \|L\| \leq \left\|\phi\right\|\right\}$. Then, there is a net $(u_i)_i \subseteq C(X)$ with $\|u_i\| \leq \left\|\phi\right\|$ for all i and

$$\int \phi \, d\mu = \lim_i \int u_i \, d\mu$$

for all $\mu \in M(X)$. For any $\psi \in B_\infty(X)$, we have $\psi\mu \in M(X)$ for any $\mu \in M(X)$. In particular, we have

$$\int \phi\psi \, d\mu = \lim_i \int u_i \psi \, d\mu$$

for all $\psi \in B_\infty(X)$ and $\mu \in M(X)$, whence $\widetilde{\rho}(u_i\psi) \to \widetilde{\rho}(\phi\psi)$ in WOT. For any $\psi \in C(X)$, we have that

$$\begin{aligned}\widetilde{\rho}(\phi\psi) &= \lim_i \widetilde{\rho}(u_i\psi) \\ &= \lim_i \rho(u_i)\rho(\psi) \\ &= \widetilde{\rho}(\phi)\rho(\psi).\end{aligned}$$

In particular, we have $\widetilde{\rho}(u_i\psi) = \rho(u_i)\widetilde{\rho}(\psi)$ for any $\psi \in B_\infty(X)$ for all $u_i \in C(X)$. Therefore, we have $\widetilde{\rho}(u_i\psi) \to \widetilde{\rho}(\phi\psi)$ in WOT, whence

$$\widetilde{\rho}(\phi\psi) = \widetilde{\rho}(\phi)\widetilde{\rho}(\psi)$$

for any $\phi, \psi \in B_\infty(X)$.

The linearity of measures gives that $\widetilde{\rho}$ itself is linear. A similar bootstrapping argument gives $\widetilde{\rho}(\phi)^* = \widetilde{\rho}\left(\overline{\phi}\right)$.

Now, let $\Delta \subseteq X$ be Borel. Let $E(\Delta) = \widetilde{\rho}(\chi_\Delta)$. We will show that E is a spectral measure.

First, since χ_Δ is a hermitian idempotent in $B_\infty(X)$, it follows that $E(\Delta)$ is a projection by the classification of projections on a Hilbert space. Furthermore, $\chi_\emptyset = 0$ and $\chi_X = 1$, so that $E(\emptyset) = 0$ and $E(X) = 1$. Since

$$E(\Delta_1 \cap \Delta_2) = \widetilde{\rho}\left(\chi_{\Delta_1 \cap \Delta_2}\right)$$

$$= \widetilde{\rho}\left(\chi_{\Delta_1}\chi_{\Delta_2}\right)$$
$$= E(\Delta_1)E(\Delta_2).$$

Finally, let $\{\Delta_n : n \geq 0\}$ be a pairwise disjoint sequence of Borel sets. Then, for fixed $h \in H$ and $\Lambda_n = \bigcup_{k=n+1}^{\infty} \Delta_k$, we have

$$\begin{aligned}\left\| E\left(\bigcup_{k=1}^{\infty}\Delta_k\right) - \sum_{k=1}^{n} E(\Delta_k)\right\|^2 &= \langle E(\Lambda_n)h, E(\Lambda_n)h\rangle \\ &= \langle E(\Lambda_n)h, h\rangle \\ &= \int \chi_{\Lambda_n}\, d\mu_{h,h} \\ &= \sum_{k=n+1}^{\infty} \mu_{h,h}(\Delta_k) \\ &\xrightarrow{n\to\infty} 0.\end{aligned}$$

Thus, E is a spectral measure.

Finally, we must show that $\rho(u) = \int u\, dE$ for all $u \in C(X)$. In fact, we will show that $\widetilde{\rho}(\phi) = \int \phi\, dE$ for all $\phi \in B_\infty(X)$. For a fixed $\phi \in B_\infty(X)$, we choose a Borel partition $\{\Delta_1, \ldots, \Delta_n\}$ such that $\sup\left\{\left|\phi(x) - \phi(x')\right| : x, x' \in \Delta_k\right\} < \varepsilon$ for each k. Then,

$$\left\| \phi - \sum_{k=1}^{n} \phi(x_k)\chi_{\Delta_k} \right\| < \varepsilon$$

for any choices of $x_k \in \Delta_k$. Since $\left\|\widetilde{\rho}\right\| = 1$, it follows that

$$\left\| \widetilde{\rho}(\phi) - \sum_{k=1}^{n} \phi(x_k)E(\Delta_k) \right\| < \varepsilon,$$

so that $\widetilde{\rho}(\phi) = \int \phi\, dE$ for any $\phi \in B_\infty(X)$.

The uniqueness of E follows from the fact that the measures $E_{g,h}$ are unique for each $g, h \in H$ as part of the Riesz Representation Theorem.

Remark: By the way the measures $E_{g,h}$ are defined (i.e., they are regular Borel measures by the Riesz Representation Theorem), it follows that the spectral projections $E(U)$ are nonzero when evaluated on any open subset of $\sigma(N)$.

The primary use of the existence of spectral measures is to enable the Borel functional calculus for normal operators on Hilbert spaces.

Theorem: Let N be a normal operator on H. There is a unique spectral measure E on the Borel subsets of $\sigma(N)$ such that:

(a) $N = \int z\, dE$;

(b) if G is a nonempty open subset of $\sigma(N)$, then $E(G) \neq 0$;

(c) if $T \in B(H)$, then $TN = NT$ and $TN^* = N^*T$ if and only if $TE(\Delta) = E(\Delta)T$ for all Borel $\Delta \subseteq \sigma(N)$.

Proof. Let $A = C^*(N)$. Then, A is the closure of the polynomials in N and N^*, so there is an isometric $*$-isomorphism $\rho\colon C(\sigma(N)) \to A$ given by the functional calculus, $\rho(u) = u(N)$. There is a unique spectral measure E defined on the Borel subsets of $\sigma(N)$ such that $\rho(u) = \int u\, dE$ for all $u \in C(\sigma(N))$. This gives (a).

If G is a nonempty open subset of $\sigma(N)$, Urysohn's Lemma yields a continuous function u on $\sigma(N)$ such that

$0 \le u(z) \le \chi_G(z)$ for all $z \in \sigma(N)$. Since $E(G) = \widetilde{\rho}(\chi_G) \ge \rho(u) \ne 0$, it follows that $E(G) \ne 0$.

Let $T \in B(H)$ be such that $TN = NT$ and $TN^* = N^*T$. Then, by the Stone–Weierstrass theorem, it follows that $Tu(N) = u(N)T$ for all $u \in C(\sigma(N))$.

Consider the set Ω defined as follows:

$$\Omega = \{\Delta \subseteq \sigma(N) : \Delta \text{ is a Borel subset and } AE(\Delta) = E(\Delta)A\}.$$

Notice that by the definition of the spectral measure, Ω is closed under finite intersections, countable unions (using SOT convergence), and contains $\sigma(N)$, so it is a σ-algebra by the Dynkin π-λ theorem. If $G \subseteq \sigma(N)$ is open, there is a sequence $(u_n)_n$ of positive continuous functions on $\sigma(N)$ such that $u_n(z) \nearrow \chi_G(z)$ for all $z \in \sigma(N)$. Dominated convergence then gives

$$\begin{aligned}
\langle AE(G)g, h\rangle &= \langle E(G)g, A^*h\rangle \\
&= E_{g,A^*h}(G) \\
&= \int \chi_G(z)\, dE_{g,A^*h} \\
&= \int \lim_{n\to\infty} u_n(z)\, dE_{g,A^*h} \\
&= \lim_{n\to\infty} \int u_n(z)\, dE_{g,A^*h} \\
&= \lim_{n\to\infty} \langle Au_n(N)g, h\rangle \\
&= \lim_{n\to\infty} \langle u_n(N)Ag, h\rangle \\
&= \langle E(G)Ag, h\rangle.
\end{aligned}$$

Thus, Ω contains all the open subsets of $\sigma(N)$, so it must be the collection of Borel sets. The converse follows from bootstrapping by approximating with simple functions. □

The spectral measure will provide us a way to reframe normal operators as multiplication operators on $L^\infty(X, \mu)$ for some measure space (X, μ).

Definition: We call $e_0 \in H$ *star-cyclic* for an operator A if the smallest reducing subspace for A that contains e_0 is H.

The operator A is star-cyclic if it contains a star-cyclic vector.

Similarly, $e_0 \in H$ is cyclic for A if H is the smallest invariant subspace for A that contains e_0, and A is cyclic if it contains a cyclic vector.

Equivalently, e_0 is star-cyclic if and only if

$$H = \overline{\{Te_0 : T \in C^*(A)\}}$$

and e_0 is cyclic if and only if

$$H = \overline{\{p(A)e_0 : p \in \mathbb{C}[x]\}}.$$

Proposition: If A has either a cyclic or star-cyclic vector, then H is separable.

Proof. Notice that $C^*(A)$ and $\{p(A) : p \in \mathbb{C}[x]\}$ are separable subalgebras of $B(H)$. □

If μ is a compactly supported measure on $K \subseteq \mathbb{C}$, and N_μ is the operator defined by $N_\mu \equiv M_z$, where $(M_z f)(z) = zf(z)$, then we observe that $C^*\left(N_\mu\right)$ is given by M_u for all $u \in C(K)$. Since continuous functions are dense in $L^2(K, \mu)$, it follows that 1 is a star-cyclic vector for N_μ.

Furthermore, the converse holds.

Theorem: A normal operator N is star-cyclic if and only if N is unitarily equivalent to N_μ for some compactly supported measure μ on $\mathbb{C}$. If e_0 is a star-cyclic vector for N, then μ can be chosen such that there is an isomorphism $V : H \to L^2(K, \mu)$ with $Ve_0 = 1$ and $VNV^{-1} = N_\mu$. Furthermore, V is unique.

Proof. Suppose $N \cong N_\mu$. Then, we have already seen that N is star-cyclic.

Suppose N has a star-cyclic vector e_0. Let E be the spectral measure for N. Set $\mu(\Delta) = \langle E(\Delta)e_0, e_0\rangle = \|E(\Delta)e_0\|^2$ for each Borel subset Δ of $\mathbb{C}$.

Let $K = \operatorname{supp}(\mu)$. For any $\phi \in B_\infty(K)$, we have

$$\begin{aligned}\left\|\phi(N)e_0\right\|^2 &= \left\langle |\phi|^2(N)e_0, e_0\right\rangle \\ &= \int |\phi(z)|^2 \, d\langle E(z)e_0, e_0\rangle \\ &= \int |\phi|^2 \, d\mu.\end{aligned}$$

Viewing $B_\infty(K) \subseteq L^2(K, \mu)$, we have that $U\phi = \phi(N)e_0$ defines an isometry from $B_\infty(K)$ onto $\{\phi(N)e_0 : \phi \in B_\infty(K)\}$.

Since e_0 is star-cyclic, it follows that the range of U is dense in H, so U extends to an isomorphism $U : L^2(K, \mu) \to H$.

For any $\phi \in B(K)$, then we have

$$\begin{aligned}UN_\mu U^{-1}(\phi(N)e_0) &= UN_\mu(\phi) \\ &= U(z\phi) \\ &= N\phi(N)e_0,\end{aligned}$$

so that $UN_\mu U^{-1} = N$ on the dense subset $\{\phi(N)e_0 : \phi \in B_\infty(K)\}$ of $L^2(K, \mu)$. Therefore $UN_\mu U^{-1} = N$, and $V = U^{-1}$ is our desired selection. □

We say two measures μ_1 and μ_2 are mutually absolutely continuous (or of the same measure class) if they have the same null sets. We will write $[\mu_1] = [\mu_2]$. If the respective Radon–Nikodym derivatives $\frac{d\mu_1}{d\mu_2}$ and $\frac{d\mu_2}{d\mu_1}$ are essentially bounded, then we say μ_1 and μ_2 are boundedly mutually absolutely continuous.

Theorem: We have that $N_{\mu_1} \cong N_{\mu_2}$ if and only if $[\mu_1] = [\mu_2]$.

Proof. Suppose $[\mu_1] = [\mu_2]$, and let $\phi = \frac{d\mu_1}{d\mu_2}$. If $g \in L^1(\mu_1)$, then $g\phi \in L^1(\mu_2)$, with

$$\int g\phi \, d\mu_2 = \int \phi \, d\mu_1.$$

For any $f \in L^2(\mu_1)$, we have $f\sqrt{\phi} \in L^2(\mu_2)$ with

$$\left\|f\sqrt{\phi}\right\|_{L^2(\mu_2)} = \|f\|_{L^2(\mu_1)}.$$

Define $U : L^2(\mu_1) \to L^2(\mu_2)$ by $Uf = f\sqrt{\phi}$. This is an isometry.

For any $g \in L^2(\mu_2)$, we have that $f = \phi^{-1/2}g \in L^2(\mu_1)$ with $Uf = g$, so U is surjective with $U^{-1}g = \phi^{-1/2}g$ for any $g \in L^2(\mu_2)$. For any $g \in L^2(\mu_2)$, $UN_{\mu_1}U^{-1}g = zg$, so $UN_{\mu_1}U^{-1} = N_{\mu_2}$.

Now, suppose $V : L^2(\mu_1) \to L^2(\mu_2)$ is an isomorphism satisfying $VN_{\mu_1}V^{-1} = N_{\mu_2}$. Let $\psi = V(1)$, with $\psi \in L^2(\mu_2)$. We will write $N_j = N_{\mu_j}$. Notice that for any polynomial $p \in \mathbb{C}[z, \overline{z}]$, we have $Vp\left(N_1, N_1^*\right)V^{-1} = p\left(N_2, N_2^*\right)$. Since $N_1 \cong N_2$, $\sigma(N_1) = \sigma(N_2)$, so μ_1 and μ_2 have the same support, which we will dub K.

Taking uniform limits of polynomials yields for all $u \in C(K)$, we have $Vu(N_1)V^{-1} = u(N_2)$, so we have

$V(u) = u\psi$. Since V is an isometry, we have

$$\int |u|^2 \, d\mu_1 = \int |u|^2 \left|\psi\right|^2 d\mu_2$$

for every $u \in C(K)$. Therefore, we have

$$\int v \, d\mu_1 = \int v \left|\psi\right|^2 d\mu_2$$

for all $v \in C(K)$ with $v \geq 0$. By the Riesz representation theorem, we have that $\mu_1 = \left|\psi\right|^2 \mu_2$, so $\mu_1 \ll \mu_2$. Reversing the roles of N_1 and N_2, and substituting V^{-1} for V where necessary, we obtain that $\mu_2 \ll \mu_1$, whence $[\mu_1] = [\mu_2]$. □

It is important to note that the Spectral Theorem is a purely operator-algebraic notion. That is, the notions herein may be generalized to any representation $\pi \colon C(\Sigma) \to B(H)$, yielding a spectral measure E on the Borel sets of Σ such that

$$\pi(f) = \int f \, dE.$$

When $\Sigma = \Sigma(A)$ is the character space of a (unital) C^*-algebra A, then we note that any $a \in A$ corresponds to a unique element $\hat{a} \in C(\Sigma(A))$, whence given a representation $\theta \colon A \to B(H)$, we have that

$$\pi(a) = \int \hat{a} \, dE.$$

Applications of the Spectral Theorem

Proposition: If N is a normal operator with spectral decomposition $N = \int z \, dE(z)$, we have that N is compact if and only if for every $\varepsilon > 0$, $E(\{z : |z| > \varepsilon\})$ is finite rank.

Proof. Let $\varepsilon > 0$, and let $\Delta_\varepsilon = \{z : |z| > \varepsilon\}$, with $E_\varepsilon = E(\Delta_\varepsilon)$. Then,

$$\begin{aligned} N - NE_\varepsilon &= \int z \chi_{\Delta_\varepsilon^c} \, dE(z) \\ &= \phi(N), \end{aligned}$$

where $\phi(z) = z\chi_{\Delta_\varepsilon^c}(z)$. In particular, $\|N - NE_\varepsilon\| \leq \varepsilon$. If E_ε has finite rank for every $\varepsilon > 0$, then NE_ε has finite rank, so $N \in \overline{F(H)}$.

If N is compact, then $\phi(z) = z^{-1}\chi_{\Delta_\varepsilon}(z)$ has $\phi(z) \in B(\mathbb{C})$. Since N is compact, we have that $N\phi(N)$ is compact, but then

$$\begin{aligned} N\phi(N) &= \int z z^{-1} \chi_{\Delta_\varepsilon}(z) \, dE(z) \\ &= E_\varepsilon, \end{aligned}$$

so E_ε is a compact projection, whence it is finite rank. □

Theorem: If H is separable, and I is an ideal of $B(H)$ that contains a noncompact operator, then $I = B(H)$.

Proof. Let $A \in I \setminus K(H)$. Take the spectral decomposition

$$A^*A = \int t \, dE(t).$$

By the preceding proposition, there is some $\varepsilon > 0$ such that $P = E(\{t \geq 0 : \varepsilon < t < \infty\})$ has infinite rank. This

yields that

$$P = \left(\int t^{-1} \chi_{(\varepsilon,\infty)}(t)\, dE(t) \right) A^*A \in I.$$

Since H is separable, $\operatorname{hdim}(PH) = \aleph_0 = \operatorname{hdim}(H)$. There is a surjective isometry $U : H \to PH$ satisfying $1 = U^*PU$. Sine $P \in I$, $1 \in I$, so $I = B(H)$. □

Corollary: The only nontrivial closed ideal of $B(H)$ is $K(H)$.

Proposition: If N is a normal operator on H, then there are reducing subspaces H_i such that $H = \bigoplus_{i\in I} H_i$ is a direct sum of star-cyclic subspaces.

Proof. Use Zorn's lemma to find a maximal set of vectors $\mathcal{E}$ such that, for any $e, f \in \mathcal{E}$ with $e \neq f$, we have that

$$H_e = \overline{\{p(T,T^*)e : p \in \mathbb{C}[z,\overline{z}]\}}$$
$$H_f = \overline{\{p(T,T^*)f : p \in \mathbb{C}[z,\overline{z}]\}}$$

are orthogonal. Then, we have that $H = \bigoplus_{e\in\mathcal{E}} H_e$. □

Corollary: Every normal operator is unitarily equivalent to the direct sum of star-cyclic normal operators.

Theorem: If N is a normal operator on H, then there is a measure space (X, Ω, μ) and a function $\phi \in L^\infty(X,\mu)$ such that $N \cong M_\phi$ on $L^2(X,\mu)$.

Proof. If $\{N_i\}_{i\in I}$ is a collection of star-cyclic normal operators such that $N \cong \bigoplus_{i\in I} N_i$, then $\sigma(N_i) \subseteq \sigma(N)$ for each N_i.

Let (X_i, Ω_i, μ_i) be measure spaces with μ_i a Borel measure, $X_i = \operatorname{supp}(\mu_i)$, and $N_i \cong N_{\mu_i}$ acting on the measure space $L^2(X_i,\mu_i)$. Then, defining $X = \coprod_{i\in I} X_i$, the corresponding disjoint union σ-algebra is defined to be the collection of all $\Delta \subseteq X$ such that $\Delta \cap X_i \in \Omega_i$ for all $i \in I$. The corresponding measure is then given by

$$\mu(\Delta) = \sum_{i\in I} \mu_i(\Delta \cap X_i),$$

and (X, Ω, μ) is then a measure space. For any $f \in L^2(X,\mu)$, we have that $f_i = f|_{X_i}$ is an element of $L^2(X_i,\mu_i)$, and the map $U : L^2(X,\mu) \to \bigoplus_{i\in I} L^2(X_i,\mu_i)$ given by taking $Uf = \bigoplus_{i\in I} f|_{X_i}$ is an isomorphism.

Define $\phi : X \to \mathbb{C}$ by taking $\phi(z) = z$ if $z \in X_i$, where we observe that $X_i \subseteq \sigma(N)$ for each i. Therefore, ϕ is a bounded function. For any open subset of $\mathbb{C}$, we have that $\phi^{-1}(G) \cap X_i = G \cap X_i \in \Omega_i$, so ϕ is Ω-measurable, whence $\phi \in L^\infty(X,\mu)$.

It then follows that $UM_\phi U^{-1} = \bigoplus_{i\in I} N_{\mu_i} \cong N$. □

Proposition: If A is a von Neumann subalgebra of $B(H)$, then A is the norm-closed linear span of its projections.

Proof. For any $a \in A$, we have that $a + a^*$ and $a - a^*$ are in A, so A is the linear span of $\operatorname{Re}(A)$.

Let $a \in \operatorname{Re}(A)$, and set the spectral decomposition

$$a = \int t\, dE(t).$$

Approximating by piecewise linear functions, we have for any $[a,b] \subseteq \mathbb{R}$, there is a sequence $(u_n)_n$ of continuous functions over $\mathbb{R}$ such that $0 \leq u_n \leq 1$, $u_n(t) = 1$ for any $a \leq t \leq b - \frac{1}{n}$ and $u_n(t) = 0$ for all $t \leq a - \frac{1}{n}$ and $t \geq b$. Therefore, $u_n(t) \to \chi_{[a,b)}(t)$ as $n \to \infty$.

For any $h \in H$, we have

$$\|(u_n(a) - E([a,b)))h\| = \int \left|u_n(t) - \chi_{[a,b)}(t)\right|^2 dE_{h,h}(t),$$

which goes to zero by dominated convergence. Therefore, $u_n(a) \to E([a,b))$ in SOT. Thus, all the spectral projections of the form $E([a,b))$ are in A.

For any open interval (α,β) containing $\sigma(A)$, we have for any $\varepsilon > 0$ a partition $\mathcal{P} = \{\alpha = t_0 < \cdots < t_n = \beta\}$ such that

$$\left| t - \sum_{k=1}^{n} t_k \chi_{[t_{k-1} - t_k)} \right| < \varepsilon$$

for any $t \in \sigma(A)$. Therefore,

$$\left\| a - \sum_{k=1}^{n} E([t_{k-1}, t_k)) \right\| < \varepsilon,$$

meaning that $\mathrm{Re}(A) \subseteq \overline{\mathrm{span}}(P(A))$, yielding the desired result. □

Commutants and von Neumann Algebras

Proposition: Let $L\colon B(H) \to \mathbb{C}$ be a linear functional. The following are equivalent:

(i) L is SOT-continuous;

(ii) L is WOT-continuous;

(iii) there are $g_1,\ldots,g_n$ and $h_1,\ldots,h_n$ in H such that $L(A) = \sum_{k=1}^{n} \langle Ag_k, h_k\rangle$.

Proof. We have that (iii) implies (ii) and (ii) implies (i), so we only need to show that (i) implies (iii). Let L be SOT-continuous. Then, there are vectors $g_1,\ldots,g_n$ in H such that

$$\begin{aligned} |L(A)| &\leq \sum_{k=1}^{n} \|Ag_k\| \\ &\leq \sqrt{n}\left(\sum_{k=1}^{n} \|Ag_k\|\right)^{1/2}. \end{aligned}$$

We may thus define

$$p(A) = \sqrt{n}\left(\sum_{k=1}^{n} \|Ag_k\|^2\right)^{1/2},$$

and we have that $|L(A)| \leq p(A)$. Notice that $p(A)$ is a seminorm, and that $p(A) = 0$ implies $L(A) = 0$.

Let

$$K = \overline{\{Ag_1 \oplus \cdots \oplus Ag_n : A \in B(H)\}}.$$

In particular, $K \subseteq H^{\oplus n}$. Notice that if $Ag_1 \oplus \cdots \oplus Ag_n = 0$, then $p(A) = 0$, so $L(A) = 0$. If we define the linear functional $F(Ag_1 \oplus \cdots \oplus Ag_n) = L(A)$, then we have that F is well-defined on a dense subspace of K with

$$|F(Ag_1 \oplus \cdots \oplus Ag_n)| \leq \|Ag_1 \oplus \cdots \oplus Ag_n\|,$$

so that F can be extended to a bounded linear functional on $H^{\oplus n}$. Since all linear functionals on $H^{\oplus n}$ are given by

$$F_1(f_1 \oplus \cdots \oplus f_n) = \langle f_1 \oplus \cdots \oplus f_n, h_1 \oplus \cdots \oplus h_n\rangle$$

$$= \sum_{k=1}^{n} \langle f_k, h_k \rangle,$$

it follows that $L(A) = \sum_{k=1}^{n} \langle Ag_k, h_k \rangle$. □

Theorem: Let $M \subseteq B(H)$ be a $*$-subalgebra. A closed subspace $K \subseteq H$ is reducing for M if and only if the projection $p := P_K$ is in the commutant of M.

Proof. Suppose K is reducing for M. Then, for all $x \in M$ and $\xi \in K$, we have $x\xi \in K$, so that $x\xi = px\xi = xp\xi$. Analogously, since K is reducing if and only if K and $K^\perp$ are invariant under M, it follows that $\eta \in K^\perp$ satisfies $px\eta = 0 = xp\eta$, whence $px = xp$.

If $p \in M'$, then we have for all $\xi \in K$ and $\eta \in K^\perp$ that

$$\begin{aligned} \langle x\xi, \eta \rangle &= \langle xp\xi, \eta \rangle \\ &= \langle px\xi, \eta \rangle \\ &= \langle x\xi, p\eta \rangle \\ &= 0, \end{aligned}$$

so that $x\xi \in (K^\perp)^\perp = K$, or that both K and $K^\perp$ are M-invariant. □

Theorem (Double Commutant Theorem)**:** Let M be a unital $*$-subalgebra of $B(H)$. Then, $\overline{M}^{\text{SOT}} = \overline{M}^{\text{WOT}} = M''$.

Proof. Since M'' is a WOT-closed subalgebra, we have the inclusions $\overline{M}^{\text{SOT}} \subseteq \overline{M}^{\text{WOT}} \subseteq M''$, so we only need to show that $M'' \subseteq \overline{M}^{\text{SOT}}$.

If $x \in M''$, it suffices to show that for all $\xi_1, \dots, \xi_n \in H$ and all $\varepsilon > 0$ there is some $y \in M$ such that $\left\| (x-y)\xi_i \right\| < \varepsilon$.

Observe that

$$\left(M^{(n)}\right)' = \left\{ \left(a_{ij}\right)_{i,j} \in \mathbb{M}_n(B(H)) : a_{ij} \in M' \right\}.$$

Define $\xi = (\xi_1, \dots, \xi_n) \in H^{(n)}$. Then,

$$S = \overline{\left\{ y^{(n)}\xi : y \in M \right\}}$$

is reducing for M. Therefore, the projection $p \in \mathbb{M}_n(B(H))$ is in the commutant $\left(M^{(n)}\right)'$. Observe that for all $x \in M''$, we have

$$\begin{aligned} x^{(n)}\xi &= x^{(n)}p\xi \\ &= p^{(n)}x\xi \\ &\in S, \end{aligned}$$

so by the way that S was defined, we have some $y \in M$ such that $\left\| x^{(n)}\xi - y^{(n)}\xi \right\| < \varepsilon$, so that $\left\| (x-y)\xi_i \right\| < \varepsilon$ for all $i = 1, \dots, n$. □

The Characterization of Abelian von Neumann Algebras

Theorem: Let (X, Ω, μ) be a σ-finite measure space, and let $\phi \in L^\infty(X, \mu)$. Define $M_\phi f = \phi f$ for any $f \in L^2(X, \mu)$. If $A_\mu = \left\{ M_\phi : \phi \in L^\infty(X, \mu) \right\}$, then $A_\mu = A_\mu' = A_\mu''$.

Proof. It suffices to show that $A_\mu' \subseteq A_\mu$. We start by assuming that $\mu(X) < \infty$. Then, $1 \in L^2(X, \mu)$, so we define $\phi = A(1)$. Then, $\phi \in L^2(X, \mu)$. For any $\psi \in L^\infty(X, \mu)$, then $\psi \in L^2(X, \mu)$, and $A\psi = AM_\psi 1 = M_\psi A(1) =$

$\phi\psi$, with $\left\|\phi\psi\right\|_{L^2} \leq \|A\|\left\|\psi\right\|_{L^2}$.

If we let $\Delta_n = \left\{x \in X : \left|\phi(x)\right| \geq n\right\}$, and $\psi = \chi_{\Delta_n}$ in the preceding argument, then we have

$$\begin{aligned}\|A\|^2\mu(\Delta_n) &= \|A\|^2\left\|\psi\right\|^2 \\ &\geq \left\|\phi\psi\right\|^2 \\ &= \int_{\Delta_n}\left|\phi\right|^2 d\mu \\ &\geq n^2\mu(\Delta_n).\end{aligned}$$

Therefore, if $\mu(\Delta_n) \neq 0$, we have that $\|A\| \geq n$. Since A is bounded, $\mu(\Delta_n) = 0$ for some n, or that $\phi \in L^\infty(X,\mu)$. Yet, since $A = M_\phi$ on $L^\infty(X,\mu)$, and $L^\infty(X,\mu) \subseteq L^2(X,\mu)$ is dense, it follows that $A = M_\phi$ on $L^2(X,\mu)$.

In the general σ-finite case, we may decompose $X = \bigsqcup_{n=1}^\infty X_n$ as a countable union of disjoint subsets with finite measure, and use the same process to show that on each $L^\infty\left(X_n, \mu|_{X_n}\right)$, we have that $A = M_{\phi_n}$, where $\phi_n = \phi|_{X_n}$. □

Theorem (Fuglede's Theorem)**:** Suppose N and M are normal operators on H and K respectively, and there is some $B\colon K \to H$ satisfying $NB = BM$. Then, $N^*B = BM^*$.

Proof. We have that $p(N)B = Bp(M)$ for any polynomial $p(z)$. Since, for a fixed $z \in \mathbb{C}$, we have that $\exp(i\bar{z}N)$ and $\exp(i\bar{z}M)$ are limits of polynomials in N and M respectively. Thus, $\exp(i\bar{z}N)B = B\exp(i\bar{z}M)$ for all $z \in \mathbb{C}$, or that $B = e^{-i\bar{z}N}Be^{i\bar{z}M}$.

Since $\exp(X+Y) = \exp(X)\exp(Y)$ for any commuting operators X and Y, and N and M are normal, we have that

$$\begin{aligned}f(z) &= e^{-izN^*}Be^{izM^*} \\ &= e^{-izN^*}\left(e^{-i\bar{z}N}Be^{-i\bar{z}M}\right)e^{izM^*} \\ &= e^{-i(zN^*+\bar{z}N)}Be^{i(zM^*+\bar{z}M)}.\end{aligned}$$

For every $z \in \mathbb{C}$, we have that $zN^* + \bar{z}N$ and $zM^* = \bar{z}M$ are Hermitian operators. Since, for any Hermitian operator A and any real t, we have

$$\begin{aligned}e^{itA} &= \int_{\sigma(A)} e^{itx}\, dE(x) \\ e^{-itA} &= \int_{\sigma(A)} e^{-itx}\, dE(x) \\ &= \int_{\sigma(A)} \left(e^{itx}\right)^* dE(x) \\ &= \left(\int_{\sigma(A)} e^{itx}\, dE(x)\right)^* \\ &= \left(e^{itA}\right)^*,\end{aligned}$$

it follows that $e^{itA}\left(e^{itA}\right)^* = \int_{\sigma(A)} 1\, dE(x) = \left(e^{itA}\right)^* e^{itA}$, so that e^{itA} is then unitary. In particular, $\exp(-i(zN^* + \bar{z}N))$ and $\exp(i(zM^* + \bar{z}M))$ are unitary, so $\|f(z)\| \leq \|B\|$. Since $f\colon \mathbb{C} \to B(K,H)$ is an entire function, Liouville's Theorem gives that f is constant.

Thus,

$$\begin{aligned}0 &= f'(z) \\ &= -iN^*e^{-izN^*}Be^{izM^*} + ie^{-izN^*}BM^*e^{izM^*}.\end{aligned}$$

Setting $z = 0$ yields that $0 = iN^*B + iBM^*$, whence $N^*B = BM^*$. □

Definition: We say $e_0 \in H$ is a separating vector for a $*$-subalgebra $A \subseteq B(H)$ if $Te_0 = 0$ implies $T = 0$.

Proposition: If e_0 is a cyclic vector for A, then e_0 is separating for A'.

Proof. If $T \in A'$ satisfies $Te_0 = 0$, then for every $S \in A$, we have $TSe_0 = STe_0 = 0$. Since the closed A-orbit of e_0 is H, it follows that $T = 0$. □

Theorem: Let H be separable, and let A be an abelian C^*-algebra. The following are equivalent:

(i) $A = A'$;

(ii) A is unital, has a cyclic vector, and SOT-closed;

(iii) there is a compact metric space X, a positive Borel measure μ with $\operatorname{supp}(\mu) = X$, and an isomorphism $U : L^2(X, \mu) \to H$ such that $UA_\mu U^{-1} = A$.

Proof. Assume (i) holds. By Zorn's Lemma and separability of H, there is a maximal sequence of unit vectors $\{e_n : n \in \mathbb{N}\}$ with $\overline{Ae_n} \perp \overline{Ae_m}$. By maximality, $H = \bigoplus_{n=1}^{\infty} \overline{Ae_n}$.

Let

$$e_0 = \sum_{n=1}^{\infty} \frac{1}{2^{n/2}} e_n.$$

We have that $\|e_0\|^2 = 1$ by the pairwise orthogonality of the e_n. If P_n is the projection onto $H_n := \overline{Ae_n}$, we have that H_n is A-invariant, meaning that H_n is reducing for A. Thus, $P_n \in A' = A$. Since

$$\begin{aligned} \overline{Ae_0} &\supseteq \overline{AP_n e_0} \\ &= \overline{Ae_n} \\ &= H_n, \end{aligned}$$

it follows that $\overline{Ae_0} = H$, and e_0 is a cyclic vector for A. This shows (ii).

Now, suppose (ii) holds. Since H is separable, the ball B_A of A is WOT-metrizable and compact. Choosing a countable WOT-dense subset of B_A and letting A_1 be the C^*-algebra generated by this subset, it follows that A_1 is a separable C^*-algebra whose SOT closure is A.

Let $X = \Sigma(A_1)$. Let $\rho\colon C(X) \to A_1 \subseteq A \subseteq B(H)$ be the inverse Gelfand transform. There is then a spectral measure E such that $\rho(u) = \int u \, dE$ for all $u \in C(X)$. For any $\phi \in B_\infty(X)$, there is a net $(u_i)_{i \in I}$ in $C(X)$ converging to ϕ in the weak* topology. That is,

$$\int u_i \, d\nu \to \int \phi \, d\nu$$

for all $\nu \in M(X)$. Thus, $\rho(u_i) \to \int \phi \, dE$ in WOT, meaning that

$$\left\{ \int \phi \, dE : \phi \in B_\infty(X) \right\} \subseteq A$$

since A is SOT-closed.

Let e_0 be a cyclic vector for A. Set $\mu(\Delta) = \langle E(\Delta)e_0, e_0 \rangle$. Then,

$$\left\langle \left(\int \phi \, dE \right) e_0, e_0 \right\rangle = \int \phi \, d\mu$$

for all $\phi \in B_\infty(X)$. Viewing $B_\infty(X) \subseteq L^2(X, \mu)$ by identifying functions equal μ-a.e., we have

$$\left\| \left(\int \phi \, dE \right) e_0 \right\|^2 = \left\langle \left(\int |\phi|^2 \, dE \right) e_0, e_0 \right\rangle$$

$$= \int |\phi|^2 \, d\mu.$$

In particular, if we have $U : B_\infty(X) \to H$ defined by $U\phi = \left(\int \phi \, dE\right)e_0$, then U is a well-defined map from the dense subspace $B_\infty(X) \subseteq L^2(X,\mu)$ into H, and U is an isometry. Finally, the domain and range of U are dense. The density of the range emerges from the fact that, since $A = \overline{A_1}^{\text{WOT}}$, we have that $\overline{Ae_0}^{\text{wk}} = \overline{A_1 e_0}^{\text{wk}}$, so since weak and norm closures coincide on convex subsets, we have that $\overline{Ae_0}^{\|\cdot\|} = \overline{Ae_0}^{\text{wk}}$, giving the desired result.

For any $\phi \in B_\infty(X)$ and $\psi \in L^\infty(\mu)$, we have that

$$\begin{aligned} UM_\psi \phi &= U(\psi\phi) \\ &= \left(\int \psi\phi \, dE\right)e_0 \\ &= \left(\int \psi \, dE\right)\left(\int \phi \, dE\right)e_0 \\ &= \left(\int \psi \, dE\right)U\phi, \end{aligned}$$

so that $UM_\psi U^{-1} = \int \psi \, dE$, whence $UA_\mu U^{-1} \subseteq A$.

Yet, since $UA_\mu U^{-1}$ is a SOT-closed C^*-subalgebra of $B(H)$ that contains $UC(X)U^{-1} = A_1$, it follows that $UA_\mu U^{-1} = A$. Since A_1 is separable, X is thus metrizable. This yields (iii).

Finally, (iii) implies (i) since $UA_\mu U^{-1} = A$ implies that $UA'_\mu U^{-1} = A'$, and $A'_\mu = A_\mu$ by the previous theorem. □

In particular, we thus obtain that any abelian von Neumann algebra (on a separable Hilbert space) is given by $L^\infty(X,\mu)$ for some compact metric space X.

The L^∞-Functional Calculus

We will show now that given any normal operator N on $B(H)$, we have that

$$W^*(N) = \{N\}'' = \{\phi(N) : \phi \in B_\infty(\sigma(N))\}.$$

First, observe that the first equality follows from the Double Commutant theorem and Fuglede's Theorem. If we have spectral decomposition $N = \int z \, dE(z)$, then we observe that $\int \phi \, dE \in W^*(N)$. Since $W^*(N)$ is an abelian von Neumann algebra, it admits a cyclic and separating vector e_0. Let μ be defined on $\sigma(N)$ by

$$\begin{aligned} \mu(\Delta) &= \|E(\Delta)e_0\|^2 \\ &= \langle E(\Delta)e_0, e_0\rangle. \end{aligned}$$

Proposition: One has $\mu(\Delta) = 0$ if and only if $E(\Delta) = 0$.

Proof. If $\mu(\Delta) = 0$, then $E(\Delta)e_0 = 0$. Since $E(\Delta) = \chi_\Delta(N)$, and e_0 is separating for $W^*(N)$, it follows that $E(\Delta) = 0$. The reverse proposition follows immediately. □

Definition: A *scalar-valued spectral measure* for N is a positive Borel measure μ on $\sigma(N)$ such that $\mu(\Delta) = 0$ if and only if $E(\Delta) = 0$. That is, μ and E are "mutually absolutely continuous."

Technically, μ and E are not directly absolutely continuous due to the fact that E is projection-valued.

In particular, the earlier proposition says that scalar-valued spectral measures exist. We will show that every scalar-valued spectral measure is given by $\mu(\Delta) = \|D(\Delta)e_0\|^2$.

If $h \in H$, let $\mu_h = E_{h,h}$ and $H_h = \overline{W^*(N)h}$. That is, H_h is a reducing subspace for N that contains h. Let $N_h = N|_{H_h}$. Then, N_h is a $*$-cyclic normal operator with $*$-cyclic vector h. By the uniqueness of the spectral

measure for a normal operator, we have that the spectral measure for N_h is $E(\Delta)|_{H_h}$, or $\chi_\Delta(N_h) = \chi_\Delta(N)|_{H_h}$. So, there is a unique isomorphism $U_h \colon H_h \to L^2(\mu_h)$ such that $U_h h = 1$ and $U_h N_h U_h^* = zf$ for all $f \in L^2(\mu_h)$.

We already know how to analyze the local behavior of N on each of its $*$-cyclic subspaces by our earlier analysis. Our goal is to show that if h is a separating vector for $W^*(N)$, then the functional calculus for N is determined by the functional calculus for N_h. This is done by establishing a $*$-isomorphism $W^*(N) \to W^*(N_h)$ when h is a separating vector taking $A \mapsto A|_{H_h}$.

Lemma: If $h \in H$, and $\rho_h \colon W^*(N) \to W^*(N_h)$ is given by $\rho_h(A) = A|_{H_h}$, then ρ_h is a surjective WOT-continuous $*$-homomorphism. Moreover, if $\psi \in B_\infty(\sigma(N))$, then $\rho_h(\psi(N)) = \psi(N_h)$, and if $A \in W^*(N)$, then there is some $\phi \in B_\infty(\sigma(N_h))$ such that $\rho_h(A) = \phi(N_h)$.

Proof. We start by showing that $\rho_h \colon W^*(N) \to W^*(N_h)$ has the correct codomain. Observe that if $p(z, \overline{z})$ is a polynomial in z and $\overline{z}$, then $\rho_h(p(N, N^*)) = p\left(N_h, N_h^*\right)$.

If $(p_i)_i$ is a net of polynomials converging to A in WOT, then for any $f, g \in H_h$, we have

$$\langle p_i(N, N^*)f, g\rangle \to \langle Af, g\rangle$$

In particular, this means that $p_i\left(N_h, N_h^*\right) \to \rho_h(A)$ in WOT. Since ρ_h is a $*$-homomorphism as H_h is reducing for $W^*(N)$, we have the desired result.

Let $\psi \in B_\infty(\sigma(N))$. Then, there is a net of polynomials $(p_i(z, \overline{z}))_i$ such that

$$\int p_i \, d\nu \to \int \psi \, d\nu$$

for all $\nu \in M(\sigma(N))$, as emerges from the fact that $L_\psi(\nu) = \int \psi \, d\nu$ defines a bounded linear functional on $M(\sigma(N))$, and $C(K) \subseteq C(K)^{**}$ is weak*-dense. Since $\sigma(N_h) \subseteq \sigma(N)$ as $N = N_h \oplus N'$ for some other operator N', it follows that

$$\int p_i \, d\eta \to \int \psi \, d\eta$$

for every η in $M(\sigma(N_h))$. In particular, since this holds for all the measure $E_{g,h}$, it follows that $p_i(N, N^*) \to \psi(N)$ in WOT, and $p_i\left(N_h, N_h^*\right) \to \psi(N_h)$ in WOT. Yet, we have

$$\rho_h(p_i(N, N^*)) = p_i\left(N_h, N_h^*\right),$$

and $\rho_h(p_i(N, N^*)) \to \rho(\psi(N))$. Thus, $\rho_h(\psi(N)) = \psi(N_h)$.

Let $U_h \colon H_h \to L^2(\mu_h)$ is the isomorphism with $U_h h = 1$ and $U_h N_h U_h^{-1} = N_{\mu_h}$. Then, if $A \in W^*(N)$ with $A_h = \rho_h(A)$, we have $A_h N_h = N_h A_h$, so that $U_h A_h U_h^{-1} \in \left\{N_{\mu_h}\right\}'$. Thus, there is ϕ such that $U_h A_h U_h^* = M_\phi$, so that $A_h = \phi(N_h)$.

To show that ρ_h is surjective, we note that if $B \in W^*(N_h)$, then we have $B = \psi(N_h)$ for some $\psi \in B_\infty(\sigma(N_h))$ by the analogous previous argument. We may extend ψ to $\sigma(N)$ by letting $\psi = 0$ on $\sigma(N) \setminus \sigma(N_h)$. Thus, $\psi(N) \in W^*(N)$ with $\rho_h(\psi(N)) = \psi(N_h) = B$. □

Lemma: If $e \in H$ is such that μ_e is a scalar-valued spectral measure for N, and ν is a positive measure on $\sigma(N)$ such that $\nu \ll \mu_e$, then there is $h \in H_e$ such that $\nu = \mu_h$.

Proof. We let $f = \left(\frac{d\nu}{d\mu}\right)^{1/2} \in L^2(\mu_e)$, and set $h = U_e^{-1} f$. Then, $h \in H_e$, and for any Borel set Δ, we have

$$\nu(\Delta) = \int \chi_\Delta \, d\nu$$

$$
\begin{aligned}
&= \int \chi_\Delta f^2 \, d\mu_e \\
&= \left\langle M_{\chi_\Delta} f, f \right\rangle \\
&= \left\langle U_e^{-1} M_{\chi_\Delta} f, U_e^{-1} f \right\rangle \\
&= \langle E(\Delta) h, h \rangle \\
&= \mu_h(\Delta).
\end{aligned}
$$

□

Lemma: We have that

$$W^*(N) = \{\phi(N) : \phi \in B_\infty(\sigma(N))\}$$

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